

Curbing fossil fuels: on the design of global reward payment funds to induce countries to reduce supply, reduce demand and expand substitutes

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The latest version can be found [here](#).

Abstract

Currently existing international environmental institutions curb fossil fuels by paying countries for reducing demand and expanding substitutes. This paper argues that it would be beneficial to also create a new and separate institution that would pay countries for reducing their fossil fuel supply. I compare two architectures. In the first, these institutions would be separate so that donors could flexibly earmark their donations. From this, a time consistency problem arises, resulting in a version of the Weak Green Paradox. However, the time consistency problem cannot lead to a Strong Green Paradox under any empirically plausible assumptions. Under a second architecture, there would be a single coordinated institution committing to a path of splitting its overall available funding in the ex ante optimal way. A model with endogenous funding suggests that the first architecture would result in higher global welfare than the second. This is because it would allow fossil fuel exporters (importers) to donate to the institution paying countries for reducing fossil fuel supply (demand), thus raising (lowering) world market prices of fossil fuels. Consequently, there would be stronger incentives to contribute and thus higher funding. This effect appears likely to outweigh the efficiency losses that come from sub-optimal splitting of funding between supply reduction, demand reduction and substitute expansion.

1 Introduction

This project studies goods with global externalities. In applying the model I focus throughout the paper on the case of fossil fuels and particularly on coal.

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However, the results of the static model that I analyze in section 3 are arguably relevant for most goods with global externalities.

Specifically, I study the problem faced by a global institution having an exogenous budget which it can split between the following three approaches to curbing coal: It can pay countries to reduce coal extraction (supply reduction), to reduce energy use (demand reduction) and to expand renewables (substitute expansion).

The question arises as to how to split the budget between these three approaches. An analogous question arises for any good with global externalities. The following diagram summarizes how the world is currently answering this question for several important goods with negative global externalities (left column in brown) and goods with positive global externalities (left column in green):

good with global externalities	substitute	international mechanisms focused on supply	international mechanisms focused on demand	international mechanisms focused on substitute
fossil fuels	renewables, nuclear		Clean Development Mechanism, Green Climate Fund, Global Environment Facility, Climate Investment Fund, UN Environment Program	Clean Development Mechanism, Green Climate Fund, Global Environment Facility, Climate Investment Fund, UN Environment Program, IAEA's Nuclear Fuel Bank
goods produced on previously forested land (soy, palm oil...)	same goods produced in Savannah	REDD+ UN-REDD	timber certification schemes	
fish	bivalve aquaculture	WTO negotiations on reducing fishery subsidies		
drugs for infectious disease control	single product treatments increasing risk of drug resistance		Global Fund for AIDS, TB and Malaria	
goods for pandemic surveillance			International Health Regulations	

Throughout the paper I assume that there is complete information. Assuming that all demand and supply elasticities are finite, I prove that the optimal amount of funding allocated to each of the three approaches is always strictly positive and an increasing function of the total available budget (see corollary 1).

For the case of coal, I find based on middle of the road elasticity estimates taken from the literature, that for an exogenous budget it is optimal for the global institution to spend 70% on paying countries to reduce coal supply. This contrasts with the current way that global institutions try to curb fossil fuels. So far, all of the money has been spent on demand reduction and substitute expansion.¹

In practice, a new global institution could be set up that would define for each country reference levels of stocks of cumulative coal extraction based on business as usual scenario projections and then reward countries each year to the extent that their actual cumulative coal extraction is below the reference level for that year. This kind of scheme is already being used for rewarding countries for preserving tropical forests (Seymour and Busch (2016)). Ultimately, this paper tries to contribute to answering the question as to whether it would be best for such a scheme to be implemented by a separate institution or whether it would be better for there to a unified institution committing to split the flow of funding that it receives in the ex ante optimal way between all the different approaches to curbing fossil fuels (supply reduction, demand reduction, substitute expansion). However, until section 7 I assume that there is a single unified institution.

In section 5. I assume that the global institution receives an exogenous flow of funding. If the global institution can freely borrow against and save from this flow of funding and if it can fully commit to a path of reward payment schemes then the optimal scheme will leave world market prices unchanged (see the Dynamic Price Preservation Lemma 11). Moreover, in each period the global institution should spend strictly positive amounts on each of the three types of approaches (rewarding countries for supply reduction, demand reduction and substitute expansion. Furthermore, these optimal amounts are all strictly increasing functions of the present value of the global institution's available funding (Corollary 4).

The global institution faces a time consistency problem: If it promises to countries that it will stick to the path of reward payment schemes that is optimal assuming full commitment and if countries believe this and world market prices form accordingly, then the global institution will in the future have incentives to renege on the promised path of reward payment schemes and shift spending from the supply side to the demand side and substitute side relative to the initially promised paths of reward payment schemes.

To help assess the severity of this time consistency problem, I study the case where the global institution has no way to commit itself to future reward payment schemes at all. I focus on the case where the global institution needs

¹There are other cases where rewarding supply reduction is absent, for example in the case of drugs for infectious disease control as displayed in the diagram. However, in that case, marginal costs of production are presumably approximately constant in the long run and so the price elasticity of supply is arguably very large in the long run. It turns out that this implies that the optimal amount of spending on rewarding supply reduction is very small. Thus the model can rationalize the fact that the world is focusing on demand side contracting in this case. Some of the other empty boxes in the above diagram appear to not be rationalizable within the model I will present. But I leave it for future work to analyze the specifics of these cases.

to spend in each period the entire funding that it receives (or the case where it cannot borrow and this constraint is binding in each period). For the model with three periods I find that there will always be a Weak Green Paradox in the following sense: increasing funding available for the global institution in period 2 will increase coal extraction in period 1. However, I find that under all plausible assumptions, adding money to the global institution’s budget at any point in time will necessarily decrease aggregate damages from global warming, at least when the overall budget is small. In the terminology common in the literature: a Strong Green Paradox cannot plausibly arise.

To help mitigate the time consistency problem, the global institution could use part of its money to buy up appropriately chosen fossil fuel deposits. However, this can only ever be a partial remedy. This suggests that it could be valuable to explore and assess alternative mechanisms for mitigating the time consistency problem, a task that I leave for future research.

I now briefly put the results discussed so far into perspective, using the following diagram:

	separate institutions	unified institution
full commitment		architecture 2
no commitment	architecture 1	architecture 3

Ultimately, this paper tries to help assess the relative merits of architecture 1 and architecture 2. Under architecture 1, there are separate funds for supply reduction, demand reduction and substitute expansion. Donors can flexibly earmark monetary contributions to the fund that they prefer. Under architecture 2, there is a single unified institution that commits to splitting the flow of funding that it receives in the ex ante optimal way between the three approaches (paying countries to reduce supply, reduce demand or expand substitutes). Under architecture 2 donors cannot earmark monetary contributions.

It seems very complicated to compare architecture 2 directly to architecture 1 in a single model that takes into account both the intertemporal linkages arising from the stock nature of fossil fuels and that endogenizes funding for the global institutions. This is why in this paper I try to decompose the comparison between architecture 2 and architecture 1 into two parts with the help of the intermediate architecture 3.

In section 5 I compared architecture 2 to architecture 3. For this I assume that funding is exogenous, which allows me to focus on the commitment problem that arises due to the intertemporal linkage of fossil fuel prices. I then argue that under architecture 1 a similar time consistency problem arises as under architecture 3. Thus the results obtained about here help us assess the advantage of architecture 2 over architecture 1 that arise due to the avoidance of the time consistency problem.

To complement this analysis, I compare in section 7 architecture 1 to architecture 3 in a model with endogenous funding. To make the analysis tractable, I do not explicitly model the intertemporal linkage of fossil fuel prices. Under architecture 1, net coal exporters can raise the world market price of coal by giving money to the fund rewarding countries for reducing coal supply, whilst

coal importers can lower the world market price of coal by giving money to the fund rewarding countries for reducing coal demand. For both this increases the incentives to contribute money to the respective funds. These incentive effects lead to much larger contributions than in the case where countries can only contribute to a centralized institution that decides on the split of the budget. This increase in funding greatly outweighs the loss associated with the fact that the budget split between the funds for demand reduction and supply reduction is not optimal under architecture 1. I then argue that under architecture 2 endogenous funding would plausibly be even lower than under architecture 3. Thus the results obtained plausibly provide a lower bound for the advantage of architecture 1 over architecture 2 in terms of increased funding.

Taken together, the results presented in this paper can help us assess the relative merits of architecture 1 and 2: Under architecture 1 there will likely be substantially larger funding than under architecture 2. Section 7 suggests that there would be more than twice as much funding. On the other hand, there will be efficiency losses under architecture 2 due to the time consistency problem identified in section 5. However, this effect is unlikely to outweigh the benefit of larger overall funding.

Thus this paper arrives at the following tentative policy conclusions: A new fund should be created that would each year reward countries for having conserved more fossil fuels than they would otherwise. This fund should be separate from the existing international funds that are focused on demand reduction and substitute expansion. This would allow donors to earmark their monetary contributions to the fund of their choice, thereby increasing their incentives to contribute. If the world were to instead to rely entirely on a fund to buy up and conserve fossil fuel deposits then not as much global welfare could be achieved. However, such a deposit purchase fund could potentially be a beneficial complement to the proposed fund because it could partially mitigate a time consistency problem which arises because the ex ante optimal reward payment schemes (for supply reduction, demand reduction and substitute expansion) involve a stronger emphasis on supply reduction later in the future than what will be optimal ex post once the future arrives.

2 Related literature and contribution

2.1 Supply side vs demand side approaches to climate change mitigation

Harstad (2012) studies a model where the countries adversely affected by climate change act in a coordinated way. He finds that the coalition's best policy is to simply buy foreign deposits and conserve them.

The model presented in the current paper differs in that in it climate change mitigation happens only due to the countries' responses to the global institution's reward payment schemes. I find that for exogenous funding it is optimal for the global institution to use strictly positive amounts of money on contracts

rewarding supply reduction, demand reduction and substitute expansion. The Coasian approach of simply buying up fossil fuel deposits is never optimal in the model.

The current study tries to complement the literature on carbon leakage by drawing out its implications for the design of global institutions. Fæhn et al. (2017) analyze the problem of a country that tries to cause a given reduction in global emissions at a minimal cost for itself. For the case of Norway they find that two thirds of the emissions reductions should optimally come from supply reduction. This result bears some similarity to my result that under exogenous funding a global institution should optimally use 70% of its budget on spending on rewarding supply reduction. Collier and Venables (2015) provide further considerations in favor of focusing on supply side approaches.

I also contribute to the literature on the optimal roles of deposit purchase contracts and leasing contracts as instruments of supply side climate policy. I find that restricting supply side approaches to deposit purchase comes at a welfare cost, which echoes the results from Eichner et al. (2020), despite the difference in the settings.

A major limitation of the current study is that it only models a single fossil fuel (interpreted to be coal) and a clean substitute. Daubanes et al. (2020) highlight the importance of taking into account the substitution between coal and gas. Extending the model presented here to simultaneously include coal, oil and gas is left for future research.

2.2 The Green paradox

The Weak form of the Green Paradox is said to occur if a policy aimed at cutting carbon emissions paradoxically increases emissions in the short run (van der Ploeg and Withagen (2015)). Many papers have analyzed under what conditions this can occur. One subset of papers studies the outcomes when a global social planner optimizes global climate policy under some constraint on the set of policy instruments that are motivated by considerations of political feasibility (Edenhofer and Kalkuhl (2011), Rezai and van der Ploeg (2016)). Other papers in this literature study various unilateral climate policies by individual countries (Hoel (2013), van Long (2015)) or coalitions of countries (van der Ploeg (2016)).

The current paper revisits this question by instead analyzing the impact of a global institution with an exogenous budget that can affect countries' policies through the reward payment schemes that it offers to them. I assume full rationality and complete information, both on the part of the global institution and the countries that it rewards for supply reduction, demand reduction and substitute expansion. The global institution's objective is to maximize global welfare.

Despite this, I show that the Green Paradox can arise endogenously: I show with Corollary 9 that a version of the Weak Green Paradox arises inevitably if the global institution cannot borrow against future funding flows and if it cannot commit to the path of contracts that it offers to countries. Specifically, I

show in a three period model that increasing the global institution's budget for the second period will always increase fossil fuel extraction in the first period.

The Strong Green Paradox is said to occur if a policy aimed at cutting carbon emissions increases aggregate damages due to global warming (van der Ploeg and Withagen (2015)). Within the 3 period model analyzed in section 5 I study under what conditions the Strong Green Paradox arises. Whether this can occur will depend on the relative size of two effects: By the Weak Green Paradox explained in the previous paragraph, increasing funding in the second period will increase extraction in the first period. On the other hand, eventual cumulative emission will necessarily decrease, as I show formally in proposition 1. Thus on the one hand, overall cumulative emissions are lowered but on the other hand some of the emissions are shifted forward in time. The latter effect is commonly taken to be negative for global welfare.

I study the case where climate damages are in each period proportional to cumulative emissions but are then summed together by discounting using the market interest rate. For tractability, I focus on the situation where the global institution's budget is small in all period. I find an explicit set of formulae that determine whether a Strong Green Paradox occurs for small budgets. This condition appears to be far from being satisfied for any plausible assumptions about elasticities and price paths. These results thus suggest a reassuring prediction: The Strong Green Paradox cannot plausibly arise purely due to a failure to commit on the part of the global institution.

2.3 International Environmental Agreements

An assumption underlying this paper is that countries cannot reach the global optimum through Coasian bargaining. I do not provide an explanation for this. A large literature has provided explanations for the widespread observed failure to achieve efficiency. Dixit and Olson (2000) argue that the appropriate way to model Coasian bargaining is via a two stage game. In the first stage countries decide whether to participate and in the second stage they maximize their joint welfare. Since participation in the first stage is voluntary, there are strong free-riding incentives. The resulting pessimistic conclusions have also been found in the form of the Small Coalition Paradox in the literature on International Environmental Agreements (Barrett 2004).

An alternative explanation for the failure of the Coase Theorem is provided by Martimort and Sand-Zantman (2016) based on asymmetric information. They show that even if an agreement is such that upon withdrawal by one country all others will stop participating, the first best cannot be reached if countries' types are sufficiently heterogeneous.

The current study does not provide an explanation for the absence of Coasian bargaining. Instead, it takes it as given that the world will not achieve a full correction of global externalities.

2.4 The design of compensation funds and of mechanisms to fund them

Kornek & Edenhofer (2020) study different proposed designs for compensation funds for global public good provision. They endogenize the financing of such compensation funds and find that their design greatly affects the amount of funding that will be raised for them. Thus they highlight that the questions of how to best design compensation funds and how to best design mechanisms to fund them are inextricably linked. I arrive at a similar conclusion in section 7.

3 The static model

The set of countries is denoted I . Each country is assumed to be of negligible size so that it acts as a price taker on the world market. z_i is the amount of energy from renewables that country i produces. x_i denotes the amount of coal that country i extracts. Coal is measured so that one unit of coal generates one unit energy via combustion. The energy generated from coal is assumed to be a perfect substitute to the energy generated from renewables. We denote by y_i the amount of energy that country i uses. All other energy sources are assumed away.

There is a common numeraire good. Its price is normalized to 1. There are no trade costs and there is a global market for coal. The world market price of coal is denoted p . Each country $i \in I$ takes world market prices as given. Each country i has energy x_i from coal and energy z_i from renewables. If the sum $x_i + z_i$ exceeds its energy use y_i then the country exports the excess amount of energy, $x_i - y_i + z_i$, in the form of coal. If the sum $x_i + z_i$ is less than its energy use y_i then the country imports the shortfall of energy, $x_i - y_i + z_i$, in the form of coal. In either case, the net revenue that the country gets is $p(x_i - y_i + z_i)$.²

Country i 's utility is quasilinear in the numeraire:

$$U_i(x_i, y_i, z_i, p) = B_i(y_i) - C_i(x_i) - G_i(z_i) + p(x_i - y_i + z_i) + f_i(x_i, y_i, z_i)$$

Here $B_i(y_i)$ is the benefit that country i derives from energy use.³ $C_i(x_i)$ is country i 's cost of extracting x_i of coal. $G_i(z_i)$ is country i 's cost of producing z_i of energy from renewables. Moreover, $f_i(x_i, y_i, z_i)$ denotes the transfer that country i gets from the global institution, as explained further below.

It will be convenient to impose that costs are strictly convex and benefits strictly concave⁴:

²Currently, 20% of all coal is traded internationally. Consistent with our assumption of a globally integrated coal market, Steckel et al. (2015) find that “in the increasingly integrated global coal market the availability of a domestic coal resource does not have a statistically significant impact on the use of coal and related emissions”.

³This should be interpreted to be the entire surplus that the country reaps from energy use, both in the form of consumer surplus from end users and producer surplus from production using energy as an input.

⁴This assumption excludes the case of constant marginal cost, which is of interest for applications like the global health examples mentioned in the introduction. However, instead of treating this case separately, we will discuss this as a limiting case as price elasticities of supply go to infinity.

Assumption 1 (strict convexity of costs and strict concavity of benefits). $C'_i(x_i) > 0, C''_i(x_i) > 0 \forall i \forall x_i, G'_i(z_i) > 0, G''_i(z_i) > 0 \forall i \forall z_i, B'_i(y_i) > 0, B''_i(y_i) < 0 \forall i \forall y_i$.

There is a global institution which evaluates global welfare as follows:

$$W = \sum_{i \in I} U_i - \eta(\sum_{j \in I} x_j)$$

The interpretation is as follows: η is a positive and strictly increasing function. $\eta(\sum_{j \in I} x_j)$ is the aggregate value of the global climate change damages due to the aggregate amount $\sum_{j \in I} x_j$ of coal combusted. U_i is the utility that country i tries to optimize. Strictly, this should include the damage due to climate change that country i suffers due to its own coal use. However, we make the simplifying assumption that country i neglects this, in line with our assumption that all countries are of negligible size.

The global institution is endowed with an exogenous budget F . It offers reward payments to countries to induce them to reduce their coal supply, their coal demand and to expand their renewables supply. The timing is as follows: First, the global institution announces transfers that it will pay to countries conditional on their choices. $f_i(x_i, y_i, z_i)$ denotes the transfer that country i will receive if it chooses (x_i, y_i, z_i) . Since countries are sovereign, the global institution cannot ask countries to pay it money, which means that the transfers $f_i(x_i, y_i, z_i)$ are constrained to be non-negative. Each country i takes the reward payment scheme $f_i(x_i, y_i, z_i)$ and the price p as given and chooses (x_i, y_i, z_i) so as to maximize its utility $U_i(x_i, y_i, z_i, p)$.

Definition 1. A reward payment scheme offered to country i is a map $f_i(x_i, y_i, z_i)$ assigning a nonnegative transfer to country i . A world market equilibrium under a given set of reward payment schemes $(f_i)_{i \in I}$ is a combination of an allocation $(x_i, y_i, z_i)_i$ and world market price p such that:

- 1) market clearing: $\sum_{i \in I} x_i - y_i + z_i = 0$
- 2) individual rationality: $(x_i, y_i, z_i) = \operatorname{argmax}_{(x,y,z)} -C_i(x) + B_i(y) - G_i(z) + p(x - y + z) + f_i(x, y, z) \forall i \in I$

Definition 2. A set $(f_i)_{i \in I}$ of reward payment schemes implements the allocation-price pair $((x_i, y_i, z_i)_{i \in I}, p)$ with a budget F if $((x_i, y_i, z_i)_{i \in I}, p)$ is a world market equilibrium under $(f_i)_{i \in I}$ and $\sum_{i \in I} f_i(x_i, y_i, z_i) = F$.

Definition 3. An reward payment scheme $f_i(x_i, y_i, z_i)$ is called **additively separable** if it can be written as $f_i(x_i, y_i, z_i) = f_{ix}(x_i) + f_{iy}(y_i) + f_{iz}(z_i)$. A reward payment scheme is called a “positive affine linear scheme” if it can be written as $f_i(x_i, y_i, z_i) = \max(0, \theta_{ixt}(\tilde{x}_{it} - x_{it})) + \max(0, \theta_{iy}(\tilde{y}_i - y_i)) + \max(0, \theta_{izt}(z_{it} - \tilde{z}_{it}))$. We will call the parameter θ_{ix} “the rate at which country i is rewarded for reducing coal extraction” and $\tilde{x}_i$ “the reference level relative to which country i is rewarded”, and similarly for the other variables.

We will now prove that (under our assumption of convex cost functions and concave benefit functions) nothing is lost by restricting attention to the positive affine linear schemes for the reward payment schemes. We will also show that we can view the global institution as if it was choosing the allocation and the world market prices.

Lemma 1 (The Surjectivity Lemma). *Consider a combination of an allocation, $(x_i, y_i, z_i)_{i \in I}$ satisfying $\sum_{i \in I} x_i - y_i + z_i = 0$ and world market price p . There exists a set $(f_i)_{i \in I}$ of positive affine linear schemes implementing $((x_i, y_i, z_i)_{i \in I}, p)$. Moreover, the minimal transfers required to implement $((x_i, y_i, z_i)_{i \in I}, p)$ under affine linear schemes are F_{ix} for rewarding country i for supply reduction, F_{iy} for rewarding country i for demand reduction and F_{iz} for rewarding country i for substitute expansion with*

$$\begin{aligned} F_{ix} &:= \sup_x px - C_i(x) - (px_i - C_i(x_i)) \\ F_{iy} &:= \sup_y B_i(y) - py - (B_i(y) - py) \\ F_{iz} &:= \sup_z pz - G_i(z) - (pz - G_i(y)) \end{aligned}$$

Moreover, there does not exist any set of reward payment schemes implementing $((x_i, y_i, z_i)_{i \in I}, p)$ with a strictly smaller budget, i.e. with a budget strictly less than $\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}$. Furthermore, these minimal required transfers are the same if we allow for any reward payment schemes (instead of restricting them to be positive affine linear).

Proof. See Appendix A.1 □

Let us now summarize the Lemmas using the diagram below. In practice, the global institution chooses reward payment schemes. The notion of market equilibrium defined above yields a mapping assigning a combination of world market price and an allocation to each set of reward payment schemes. By the surjectivity Lemma 1, this map is surjective.

The surjectivity of this market equilibrium map allows us to view the global institution as if it was choosing a combination of world market price and an allocation. Given world market price and an allocation, there are many reward payment schemes inducing them via the market equilibrium map. The transfers that end up being paid are $f_{xi}(x_i), f_{yi}(y_i), f_{zi}(z_i)$. The minimal required transfers F_{ix}, F_{iy}, F_{iz} are given by the formulae shown below:

$$\begin{array}{ccc} & \text{market equilibrium} & \\ & \text{(surjective map)} & \\ (f_{xi}, f_{yi}, f_{zi})_{i \in I} & \longrightarrow & (p, (x_i, y_i, z_i)_{i \in I}) \\ & & \downarrow \text{minimal required} \\ & & \text{transfers} \\ & & (F_{ix}, F_{iy}, F_{iz}) \\ & & F_{ix} = \sup_x px - C_i(x) - (px_i - C_i(x_i)) \\ & & F_{iy} = \sup_y B_i(y) - py - (B_i(x_i) - px_i) \\ & & F_{iz} = \sup_z pz - G_i(z) - (pz_i - G_i(z_i)) \end{array}$$

In this diagram we are implicitly restricting attention to additively separable reward payment schemes. This is justified by the Surjectivity Lemma : This

restriction does not affect the minimal transfers that are required.⁵

Hence the global institution's problem can be written as:

$\max_{(p, (x_i, y_i, z_i)_{i \in I})} \sum_{j \in I} U_j - \eta(\sum_{j \in I} x_j)$ subject to the market clearing constraint, $\sum_{i \in I} x_i - y_i + z_i = 0$, and the budget balance constraint that $\sum_{j \in I} F_{jx} + F_{jy} + F_{jz} \leq F$, where F is the exogenous budget at the global institution's disposal.

Lemma 2 (The First Best). *For a sufficiently large budget F , the first best is characterized by the following conditions plus the feasibility condition $\sum_{i \in I} (x_i - y_i + z_i)$:*

$$B'_i(y_i) = B'_j(y_j) = G'_i(z_i) = G'_j(z_j) = C'_i(x_i) + \eta = C'_j(x_j) + \eta \forall i, j \in I$$

There exists a continuum of ways to split the budget between supply reduction reward payments schemes on the one hand and demand reduction and substitute expansion reward payment schemes on the other, all of which can achieve the optimal global welfare.

Proof. See appendix A.2. □

Lemma 2 asserts that if the global institution's budget constraint is not binding, then it does not matter how exactly the budget is split between the different approaches, as long as the resulting required budget does not exceed the available budget. However, from now on we will assume that the global institution's budget constraint *is* binding:

Assumption 2. *The global institution's budget constraint is binding. In other words: its budget is insufficient to fully correct the global externalities from coal.*

This assumption will mean that it *will* matter for global welfare how the global institution's budget is split, thereby overturning the conclusion from Lemma 2. This is because the world market price of coal affects the sizes of the transfers required to make countries change their actions instead of just ignoring the reward payments. The world market price of coal, in turn, is affected by how countries are rewarded: The stronger the reward payments for supply reduction, the weaker the supply of coal on the world market and therefore the lower the resulting world market price. On the other hand, the stronger the reward payments for demand reduction and substitute expansion, the lower the demand for coal on the world market and thus the lower the resulting world market price for coal.

The following Lemma states that it is always optimal to choose a mixture of these two kinds of approaches, balanced precisely such that the net effect of the world market price of coal is neutral. It is important to emphasize that this "Price Preservation Lemma" refers to the *world market price* and not to the *net prices* that actors will face. Within a given country, the price that actors will face is the sum of the world market price and any taxes (or regulation-induced

⁵Interestingly, this no longer holds if we were to depart from the assumption of complete information. In fact, one can deduce from the results in Armstrong and Rochet (1999) that even if types are independent across dimensions the optimal mechanism will not be additively separable.

carbon prices, etc.) that the government will set. When the global institution rewards countries for supply reduction then this effectively means that it will pay countries for setting a carbon price on the coal extracted on its territory. When the global institution rewards countries for demand reduction then this effectively means that it will pay countries for taxing energy use (by households and firms). When the global institution rewards countries for expanding renewables it pays countries for subsidizing renewables.

The result of all this will always be that the *net price* of coal combustion will increase. What will optimally be preserved is the *world market price* of coal:

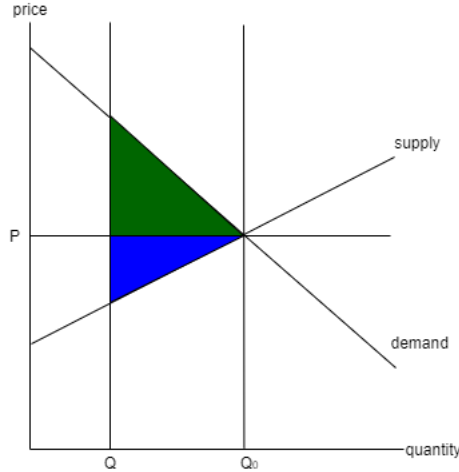
Lemma 3 (The Price Preservation Lemma, static version). “If a set of reward payment schemes achieves a given allocation with minimal aggregate transfer payments then it must preserve the world market price p of coal.”

Formally: Consider a fixed allocation $(x_i, y_i, z_i)_{i \in I}$. Consider a set of reward payment schemes $(f_i(x_i, y_i, z_i))_{i \in I}$ implementing $((x_i, y_i, z_i)_{i \in I}, p)$ for some price p under a budget F . Then if there is no other set of reward payment schemes $(\tilde{f}_i(x_i, y_i, z_i))_{i \in I}$ implementing $((x_i, y_i, z_i)_{i \in I}, \tilde{p})$ for some price $\tilde{p}$ under a budget $\tilde{F}$ with $\tilde{F} < F$ then p must equal the price of energy in the absence of any mechanism.

In particular, the optimal mechanism must leave the world market price p of coal at the same level as when there are no reward payment schemes.

Proof. See Appendix A.3 □

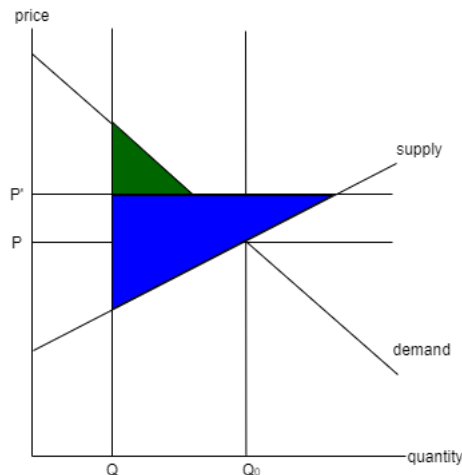
For the case where there is no substitute, we can illustrate the Price Preservation Lemma graphically. Suppose for simplicity that all countries have identical demand and supply functions shown in the following diagram:



In the absence of any mechanism, the world market price for coal is p and the quantity of coal produced and used by each country is Q_0 . Now let us compare different ways of reducing the quantity produced (and used) to $Q < Q_0$. Suppose first the global institution achieves this using a mechanism that leaves the price

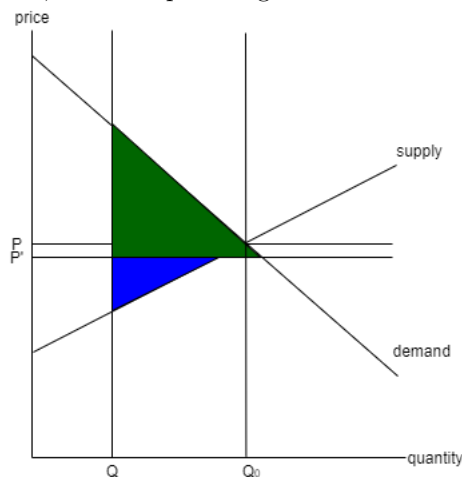
unchanged. In that case, the minimal transfer that it has to pay each country for reducing their coal use from Q_0 to Q is given by the green area. The minimal transfer that it has to pay each country for reducing their coal extraction is given by the blue area.

Now suppose the global institution were to implement Q with a higher world market price $P' > P$. This corresponds to a higher spending on rewarding supply reduction:



We see that the total size of the green and the blue areas together is larger now than when the price was preserved at P .

Similarly, greater demand side emphasis, corresponding to a smaller price $P' < P$, would require larger overall transfers:



Thus we have graphically recovered the Price Preservation Lemma: If a mechanism is to achieve a given allocation with minimal aggregate transfers then the world market price p of coal must be the same as in the absence of any mechanism.

Having established the Price Preservation Lemma by holding the allocation constant and finding the price that minimizes the required transfers, let us now hold the price constant and find the optimal allocation for the given price.

Lemma 4 (The Constrained Efficiency Lemma). *At the optimal mechanism we have: The allocation (x_i, y_i, z_i) achieves maximal welfare among all allocations satisfying market clearing and having the same value of $\sum_{i \in I} x_i$. Moreover, this constrained efficiency result even holds if we add the constraint that the world market price p of coal be any fixed value.*

Proof. See Appendix A.4 □

Given the Price Preservation Lemma and the Constrained Efficiency Lemma, it is intuitively clear that as the available budget F increases, the optimal mechanism will increase the amounts spent on each of the three approaches. To see why, we note that if we were to only expand the budget for rewarding supply reduction, then the world market price p of coal would increase, in contradiction to the Price Preservation Lemma. Similarly, if we were to only expand the budget for rewarding demand reduction and the budget for rewarding substitute expansion, then the world market price of coal would fall. Moreover, from the Constrained Efficiency Lemma it is intuitively clear that the demand side budget and the substitute side budget must both expand: restricting marginal abatement to demand reduction or substitute expansion would come at an efficiency loss. I will formally validate this conclusion by proving the following:

Corollary 1 (The Interior Solution Corollary). *For the optimal mechanism given the budget F , let $F_x(F)$ denote the amount optimally used for supply side payments and similarly $F_y(F)$ the optimal demand side budget and $F_z(F)$ the optimal substitute side budget. We have: $\frac{dF_x}{dF} > 0$, $\frac{dF_y}{dF} > 0$, $\frac{dF_z}{dF} > 0 \forall F$.*

Proof. See Appendix A.5 □

Corollary 2 (The Optimal Budget Split Corollary for Small Budgets). *For the optimal mechanism given the budget F , let $F_x(F)$ denote the amount used for supply side payments and similarly $F_y(F)$ the demand side budget and $F_z(F)$ the substitute side budget. Moreover, let us denote by $X(F)$ the aggregate coal extraction and by ϵ_x the aggregate price elasticity of supply of coal, $Y(F)$ the aggregate energy consumption and by ϵ_y the aggregate price elasticity of energy demand, by $Z(F)$ the aggregate renewable energy production and by ϵ_z the aggregate price elasticity of supply of renewable energy. We “generically” have:*

$$\begin{aligned} \lim_{F \rightarrow 0} \frac{dF_x}{dF} &= \frac{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_y}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y} \\ \lim_{F \rightarrow 0} \frac{dF_y}{dF} &= \frac{\epsilon_y}{\epsilon_z + \epsilon_y} \frac{\epsilon_x \frac{X(0)}{Y(0)}}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y} \\ \lim_{F \rightarrow 0} \frac{dF_z}{dF} &= \frac{\epsilon_z}{\epsilon_z + \epsilon_y} \frac{\epsilon_x \frac{X(0)}{Y(0)}}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y} \end{aligned}$$

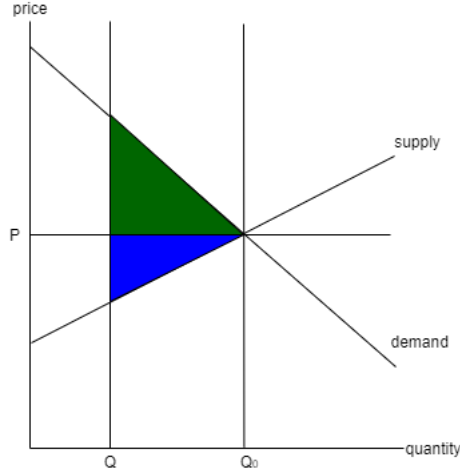
Proof. See Appendix A.5 □

We see that the more elastic the supply of coal, the smaller the proportion of money that will optimally be used to pay countries for reducing coal extraction. We even get the following:

Corollary 3. $\lim_{\epsilon_x \rightarrow \infty} \lim_{F \rightarrow 0} \frac{dF_x}{dF} = 0$

Proof. This follows directly from Corollary 2 □

For the case without any substitute we can again illustrate this result diagrammatically:



By the Price Preservation Lemma we know that the optimal mechanism consists of spending the amount corresponding to the blue area on rewarding supply reduction. Making supply more and more elastic corresponds to making the supply curve flatter and flatter. This decreases the blue area, so the optimal proportion of spending on rewarding supply reduction decreases.

We conclude this section with some concavity results:

Lemma 5. *Let $W(F)$ be the maximal welfare achievable with a given budget F . Suppose that $\eta(\sum_i x_i)$ is linear. Then $W(F)$ is strictly concave.*

Proof. See Appendix A.6 □

Lemma 6. *Let $F(F_x, W)$ denote the budget required to achieve welfare W under the further constraint that the budget spent on reducing coal supply is F_x . Then $F(F_x, W)$ is convex in F_x .*

Proof. See Appendix A.7. □

Lemmas 5 and 6 make the following conjecture plausible:

Conjecture 1. *Let $W(F_x, F_y, F_z)$ be the maximal welfare achievable under the further constraint that the amount F_x be spent on coal supply reduction, F_y be spent on energy demand reduction and F_z be spent on renewable supply expansion. $W(F_x, F_y, F_z)$ is concave.*

In the special case where all supply and demand functions have constant elasticities, conjecture 1 does seem to hold, as suggested by the numerical results shown in the next section.

4 Numerical results for the static model under constant elasticity specification

All the numerical calibrations whose results are summarized here are fully documented in the accompanying Mathematica notebook that can be downloaded [here](#).

Drawing on the literature, I use the following middle-of-the-road for the parameters ⁶:

$\epsilon_D = 0.85$ based on Freehan (2018) and Espey, J. A., & Espey, M. (2004).

$\epsilon_{S_G} = 2.7$ based on Johnson (2011)

$\epsilon_{S_C} = 1.3$ based on Dahl (2009)

$\eta = 0.5$ based on a social cost of carbon of \$100 per ton of CO₂ (based on Stern (2007))

$$\frac{X(0)}{Y(0)} = \frac{0.26}{0.26+0.4}$$

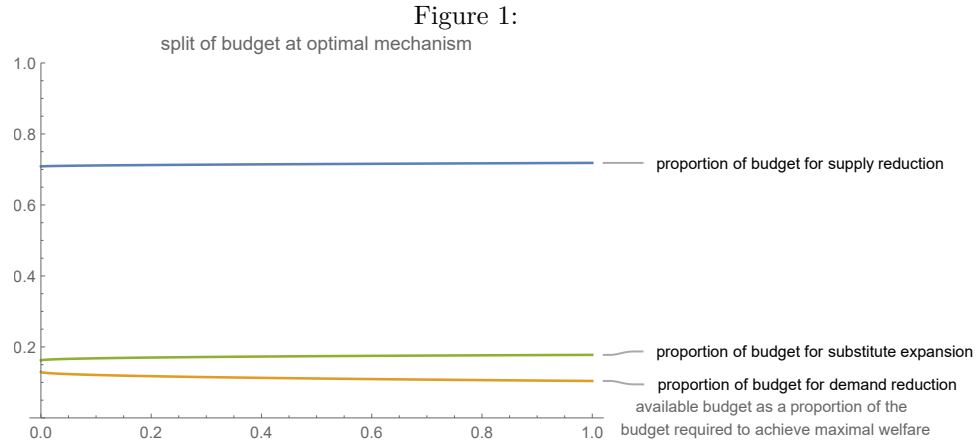
$$\frac{Z(0)}{Y(0)} = \frac{0.4}{0.26+0.4}$$

By Lemma 2, the optimal budget split for an infinitesimally small budget only depends on the elasticities at the situation where no mechanism is in place. Shown in the following figure is the optimal budget split as a function of the available budget:

⁶For the numerical calibrations that follow I use a slightly more complicated (but formally isomorphic model) that is detailed in the accompanying Mathematica notebook. The model takes into account that energy is required as an input to produce renewable energy.

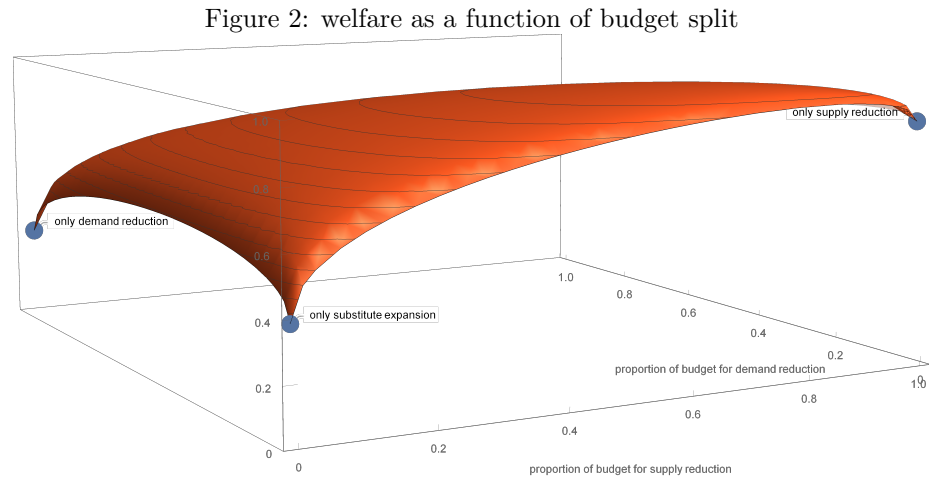
⁷Electricity prices are around \$0.2 per kwh (<https://www.iea.org/reports/world-energy-prices-2019>). Per kwh of electricity from coal 900g of CO₂ gets emitted (https://www.parliament.uk/documents/post/postpn_383-carbon-footprint-electricity-generation.pdf). Thus we have: $p = 0.2 \text{ per kwh}$ and $\eta = 0.9 \text{ kg of CO}_2 \text{ per kwh} \times \text{climate damage per ton of CO}_2 / 1000$. Assuming a social cost of carbon of \$100 per ton of CO₂, this give approximately that $\frac{\eta}{p} = 0.5$. Thus with our normalization of $p = 1$ we get $\eta = 0.5$.

⁸As of 2018, renewables generates 26 % of global electricity (<https://www.iea.org/fuels-and-technologies/renewables>), whilst coal generates 40 % of global electricity (<https://data.worldbank.org/indicator/EG.ELC.COAL.ZS>). Since the model does not take into account the other electricity sources, I ignore them for the purposes of this illustrative calibration.



The plots shows the results starting from an infinitesimal budget all the way to the minimal budget allowing the global institution to fully correct the global externality. Interestingly, the result for the optimal budget split hardly depends on the budget.

In the following figure I show how global welfare depends on how the budget is split between the three approaches:



On the x-axis and the y-axis are the proportions of the budget used for rewarding supply reduction and rewarding demand reduction, respectively. By definition, the remaining proportion of the budget is used for rewarding substitute expansion. On the z-axis is the ratio of the global welfare gains achieved divided by the maximal global welfare gains achievable with the given budget. The figure depicts the case where the budget is small. It turns out that the

results change very little when one chooses any other budget value between 0 and half the minimal value required to fully correct the global externality.

Around the optimum the surface is quite flat. In fact, as long as each of the three approaches to curbing coal is funded at at least 50% its optimal proportion, welfare losses relative to the optimal budget split are at most 12%. In C I show that this result is quite robust across the ranges of elasticity estimates found in the literature.

This result has important implications for the design of mechanisms to fund the global institution, as I will discuss in section 8. It suggests that it might not be so important to get the budget split exactly right and thus weighs in favor of decentralized funding mechanisms that have no guarantee for allocative efficiency but that create strong participation incentives by giving participating countries the opportunity to influence the allocation of funding across the different approaches to curbing fossil fuels.

5 The dynamic model

There are T discrete time periods, denoted by $t = 1, \dots, T$. Country i 's coal reserves are such that the order in which it is best to extract is always the same. Thus country i 's only relevant choice is how much to extract in each period. Country i 's cumulative extraction of coal until the end of period t is denoted by x_{it} .

Country i 's energy use in period t is denoted by y_{it} and its renewable energy production by z_{it} . Country i 's total discounted extraction cost is denoted $C_i(x_{i1}, x_{i2}, \dots, x_{iT})$, its total discounted benefit from energy consumption is $B_i(y_{i1}, y_{i2}, \dots, y_{iT})$ and its total discounted cost from renewable energy production is $G_i(z_{i1}, z_{i2}, \dots, z_{iT})$. The transfer that country i receives from the global institution in period t is denoted by $f_{it}(x_{it}, y_{it}, z_{it})$. Country i 's pursues the objective of maximizing U_i given by:

$$U_i((x_{it}, y_{it}, z_{it}, f_{it})_{t=1, \dots, T}) = -C_i(x_{i1}, x_{i2}, \dots, x_{iT}) + B_i(y_{i1}, y_{i2}, \dots, y_{iT}) - G_i(z_{i1}, z_{i2}, \dots, z_{iT}) + \sum_{t=0, \dots, T} \frac{1}{(1+r)^t} (f_{it}(x_{it}, y_{it}, z_{it}) + p_t(x_{it} + z_{it} - y_{it}))$$

Assumption 3. There exist functions b_{it}, g_{it}, c_{it} such that:

$$\begin{aligned} B_i(y_{i1}, y_{i2}, \dots, y_{iT}) &= \sum_t \frac{1}{(1+r)^t} b_{it}(y_{it}) \\ G_i(z_{i1}, z_{i2}, \dots, z_{iT}) &= \sum_t \frac{1}{(1+r)^t} g_{it}(z_{it}) \\ C_i(x_{i1}, x_{i2}, \dots, x_{iT}) &= \sum_t \frac{1}{(1+r)^t} (c_{it}(x_{it}) - c_{it}(x_{i(t-1)})) \end{aligned}$$

The form of the coal extraction cost assumed here is a discrete-time analogue of the assumption of stock-dependent extraction cost common in the continuous time literature (see e.g. van der Ploeg and Withagen (2014)). In fact, it is more general in that it allows the stock-dependent extraction cost to also explicitly depend on time.

Assumption 4. b_{it} is concave $\forall i, t$. Moreover, g_{it} and c_{it} are convex $\forall i, t$.

Analogously to the static model, the global institution evaluates global welfare as follows:

$$W = \sum_{i \in I} U_i - \eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$$

The interpretation is as follows: η represents the global climate change damages as a function of the sequence of cumulative coal combustion. U_i is the utility that country i tries to optimize. Strictly, this should include the damage due to climate change that country i suffers due to its own coal use. However, I make the simplifying assumption that country i neglects this, in line with our assumption that all countries are of negligible size. From now on I will make the following:

Assumption 5. *The global damages from climate change, $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$, is a positive function that is strictly increasing in each of its arguments.*

Lemma 7. *Assumption 5 is equivalent to the conjunction of the following two properties:*

- 1) *Additional emissions at any time increase total climate change damages.*
- 2) *Shifting emissions forward in time increases total global damages.*

Proof. See appendix A.9. □

For most of the remaining paper I do not make any further assumptions. However, for some of the results I will assume that η only depends on $\sum x_{iT}$, the eventual cumulative emissions. This assumption will greatly simplify the proofs. However, it will become clear in the proofs that this assumption can be considerably weakened without invalidating the results. I leave it for future work to identify weaker sufficient conditions for the proofs to all go through.

Definition 4. A path of reward payment schemes offered to country i is a sequence of maps $(f_{it}(x_{it}, y_{it}, z_{it}))_{t \in \{1, \dots, T\}}$, each assigning a non-negative transfer to country i . A **world market equilibrium under a given set of paths of reward payment schemes** $((f_{it})_{t \in \{1, \dots, T\}})_{i \in I}$ is a combination of an allocation $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ and a world market price path $p = (p_t)_{t \in \{1, \dots, T\}}$ such that:

- 1) market clearing: $\sum_{i \in I} x_{it} - x_{it-1} - y_{it} + z_{it} = 0 \forall t \in \{1, \dots, T\}$
- 2) individual rationality: $(x_{it}, y_{it}, z_{it})_{t \in \{1, \dots, T\}} \in \operatorname{argmax}_{(x_{it}, y_{it}, z_{it})_{t \in \{1, \dots, T\}}} - C_i(x_{i1}, x_{i2}, \dots, x_{iT}) + B_i(y_{i1}, y_{i2}, \dots, y_{iT}) - G_i(z_{i1}, z_{i2}, \dots, z_{iT}) + \sum_{t \in \{1, \dots, T\}} \frac{1}{(1+r)^t} (f_{it}(x_{it}, y_{it}, z_{it}) + p_t(x_{it} + z_{it} - y_{it}))$

From now on we will use the following assumption:

Assumption 6. $c''_{it+1}(x) < (1+r)c''_{it}(x) \forall i, t, x$.

Assumption 6 has some plausibility: If extraction technology does not change much over time then assumption 6 clearly holds.⁹

⁹In fact, assumption 6 is also plausible if extraction technology improves over time: The marginal extraction cost of coal increases as one extracts more and more of the reserves. This is because the remaining coal is harder to access. However, as time passes, technological improvements might reduce the associated increases in marginal extraction costs. This makes it plausible that we even have $c''_{it+1}(x) \leq c''_{it}(x) \forall i, t, x$.

Lemma 8. *Suppose assumption 6 holds. Then for any world market price path p there exists for each country a unique utility maximizing extraction path.*

Proof. See Appendix A.8. $\square$

Definition 5. Consider a path of world market prices $p = (p_t)_{t \in \{1, \dots, T\}}$ for coal. Let $(x_{it}^*(p))_{t \in \{1, \dots, T\}}$ denote the unique corresponding coal extraction path that maximizes country i 's utility in the absence of any reward payment schemes, assuming the world market price path p .

Lemma 9. *Suppose assumption 6 holds. Then in the absence of any reward payment schemes there is a unique world market equilibrium.*

Proof. See Appendix A.10. $\square$

Definition 6. “Affine linear reward payment schemes” are defined to be schemes where the global institution offers country i in period t the transfers $f_{iyt}(y_{it}) = \max(0, \theta_{iyt}(\tilde{y}_{it} - y_{it}))$, $f_{izt}(z_{it}) = \max(0, \theta_{izt}(z_{it} - \tilde{z}_{it}))$ and $f_{ixt}(x_{it}) = \max(0, \theta_{ixt}(\tilde{x}_{it} - x_{it}))$. We will call the parameter θ_{iyt} “the rate at which country i is rewarded for reducing energy consumption” and $\tilde{y}_{it}$ “the reference level relative to which country i is rewarded”, and similarly for the other variables.

Lemma 10 (The Surjectivity Lemma, dynamic version under full commitment). *Consider a combination of an allocation, $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ with $(x_{it})_{i \in I, t \in \{1, \dots, T\}}$ non-decreasing satisfying $\sum_{i \in I} x_{it} - y_{it} + z_{it} = 0 \forall t \in \{1, \dots, T\}$ and a world market price p . Then for any set of paths of reward payment schemes implementing this combination, the global institution must end up paying at least the amount $F_{iyt} := \sup_y b_{it}(y) - p_t y - (b_{it}(y_{it}) - p_t y_{it})$ on rewarding country i for demand reduction in period t and at least the amounts $F_{izt} := \sup_z p_t z - g_{it}(z) - (p_t z_{it} - g_{it}(z_{it}))$ on rewarding country i for substitute expansion in period t .*

Moreover, the global institution must at least spend the following amount $F_{ix} := \sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{ixt}$ in discounted money on rewarding country i for supply reduction, where F_{ixt} is defined as follows:

$$F_{ixt} := (p_t - \frac{1}{1+r} p_{t+1})(x_{it}^*(p) - x_{it}) + (c_{it}(x_{it}) - c_{it}(x_{it}^*(p))) + \frac{1}{1+r} (c_{it+1}(x_{it}^*(p)) - c_{it+1}(x_{it})) \text{ for } t \in \{1, \dots, T-1\}$$

$$F_{ixT} := p_T(x_{iT}^*(p) - x_{iT}) + (c_{iT}(x_{iT}) - c_{iT}(x_{iT}^*(p)))$$

Moreover, there does indeed exist a set of paths of reward payment schemes implementing the allocation $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ and the world market price path p that ends up paying exactly the amounts F_{iyt} on rewarding country i for demand reduction in t , the amounts F_{izt} on rewarding country i for substitute expansion in t and the discounted amount F_{ix} with i on rewarding country i for supply reduction. Furthermore, there exists a set of positive affine linear reward payment schemes achieving this.

Proof. See Appendix A.11 $\square$

A natural question is whether the dynamic version of the Surjectivity Lemma could be strengthened. Are the F_{ixt} actually the minimal amounts that the global institution has to end up paying in period t to induce country i to choose x_{it} ? It turns out that this is not true in general. To see why, we note that if in some period the global institution were to pay out sufficiently large rents to country i for sufficiently low cumulative extraction, then this could even create incentives to conserve in previous periods and thereby decrease the required transfers in those previous periods.¹⁰

5.0.1 The optimal mechanism when the global institution can fully commit and can freely save and borrow

As throughout all of the formal analysis so far (and until section 7), I assume that the funding stream available for the global institution is exogenous. Specifically, I now assume there is an exogenous flow of funding F_t being given to the global institution in period t . In this section I assume that the global institution can freely save and borrow at the interest rate r , which is also the rate at which the countries discount future money flows. Thus the global institutions has an intertemporal budget constraint: The discounted value of the aggregate transfers paid cannot exceed $F := \sum_{t \in \{1, \dots, T\}} \frac{F_t}{(1+r)^t}$.

Lemma 11 (The Price Preservation Lemma, dynamic version). *Suppose $(f_{it})_{i \in I, t \in \{1, \dots, T\}}$ is a set of reward payment schemes implementing a given allocation $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ such that the discounted value of the aggregate transfers paid to countries is minimal. Then we must have: The entire world market price path $(p_t)_{t \geq 0}$ is identical to when there is no mechanism.*

Proof. See Appendix A.12. □

Lemma 12 (The Constrained Efficiency Lemma, dynamic version with full commitment and free borrowing/saving). *Suppose the global institution has an intertemporal budget of F and it can fully commit to any path of reward payment schemes. Consider the following set of allocations $S_{F,p} := \{(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}} : \exists (f_{it})_{i,t}, p : (f_{it})_{i,t} \text{ implements } ((x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}, p) \text{ with discounted budget } F\}$*

We have: If $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ maximizes global welfare W on $S_{F,p}$ it maximizes global welfare amongst all allocations having the same value for the global damages $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$ due to climate change.

In particular we have: The allocation implemented by the optimal mechanism for a given discounted budget F is the unique allocation maximizing global welfare W under the constraint that the the global damages $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$ due to climate change be a given constant.

Proof. See Appendix A.13 □

¹⁰If we suppose that the global institution cannot commit to a path of reward payment schemes then this possibility vanishes: Even if future rents could contribute to participation incentives now, once the future arrives the global institution will no longer have incentives to pay rents. In fact, if the global institution cannot commit to a path of reward payment schemes, the Surjectivity Lemma can be strengthened, as shown in Lemma 16 below.

Intuitively, the Dynamic Interior Solution Corollary is not surprising given the Constrained Efficiency Lemma 12 : It is optimal to distribute mitigation efforts over all periods so as to minimize the discounted global welfare cost due to mitigation.

Lemma 13 (The Monotone Mitigation Lemma). *Suppose the global institution has an intertemporal budget of $F = \sum_{t \in \{1, \dots, T\}} \frac{F_t}{(1+r)^t}$ and it can fully commit to any path of reward payment schemes. Let us denote by $x_{it}(F)$ the cumulative coal extraction of i by the end of period t at the optimal mechanism. Let $\bar{F} := \sup\{F : F \text{ is insufficient to achieve full correction of externality}\}$ and suppose $F < \bar{F}$. Then we have $\frac{dx_{it}}{dF} < 0 \forall i, t$.*

Moreover, if climate change damages only depend on eventual cumulative coal emissions (i.e. if η only depends on $\sum x_{iT}$) then we also have:

$$\frac{dx_{i1}}{dF} < \dots < \frac{dx_{iT}}{dF} \text{ on } [0, \bar{F}],$$

In particular, in this case the marginal amounts of coal extracted, $\sum_i x_{it}(F) - x_{it-1}(F)$, are all strictly decreasing in F .

Proof. See Appendix A.14 □

Corollary 4 (The Dynamic Interior Solution Corollary). *Suppose that climate change damages only depend on eventual cumulative coal emissions (i.e. suppose that η only depends on $\sum x_{iT}$). Suppose the global institution has an intertemporal budget constraint (more precisely: it can freely borrow against and invest from the flow F_t of funding at a fixed and certain interest rate r) and it can fully commit to any path of reward payment schemes. Then at the optimal mechanism it will spend in each period strictly positive amounts on each of the three approaches (supply reduction, demand reduction and substitute expansion) and these amounts are strictly increasing functions of the present value $F := \sum_{t \in \{1, \dots, T\}} \frac{1}{(1+r)^t} F_t$ of the budget flow.*

Proof. By the Dynamic Price Preservation Lemma 11, world market prices must under the optimal mechanism be just as in the absence of any mechanism. By the Monotone Mitigation Lemma, both cumulative coal extraction and marginal coal extraction are strictly decreasing functions of F . The former implies that the global institution's spending on rewarding supply reduction strictly increases and the latter implies (invoking the Constrained Efficiency Lemma) that the spending on demand reduction and substitute expansion must strictly increase. □

Corollary 5 (The Coase-Does-Not-Hold Corollary). *Suppose that climate change damages only depend on eventual cumulative coal emissions (i.e. suppose that η only depends on $\sum x_{iT}$). Suppose the global institution has an intertemporal budget constraint (more precisely: it can freely borrow against and invest from the flow F_t of funding at a fixed and certain interest rate r) and it can fully commit to any path of reward payment schemes. Suppose that the global institutions restricts its supply side spending to the last period. (This would effectively be the case if the global institution were to restrict itself to buying*

up coal deposits.) Then the global institution cannot achieve optimal welfare. Thus we have: The “Coase Theorem” does not hold when the global institution’s budget is insufficient to achieve a full correction of the global externalities.

Proof. This follows directly from the Dynamic Interior Solution Corollary. $\square$

Corollary 6 (The Time Inconsistency Corollary). *Suppose the global institution announces at time 1 the optimal mechanism assuming it fully commits to it. Suppose that at time T reneges, to all countries surprise, on its promised reward payment schemes and instead offers a set of reward payment schemes that maximize global welfare, given the state $(x_{iT-1})_{i \in I}$ of the world then. Then this new reward payment scheme used for the last period involves less spending on rewarding supply reduction than the originally announced mechanism.*

Proof. See Appendix A.15. $\square$

Intuitively, corollary 6 makes sense: Part of the benefits of future spending on rewarding supply reduction are that they will by raising world market prices then increase the incentives to conserve in the prior periods. However, once the future arrives, this consideration disappears and thus the global institution is better off spending less on rewarding supply reduction.¹¹

5.1 The optimal mechanism when the global institution can commit and cannot borrow

In this section we show that the time consistency problem identified in corollary 6 persists if we add the constraint that the global institution cannot borrow against future funding flows.

Lemma 14 (The Price Rise Lemma). *Suppose $T = 2$. Suppose that the no-borrowing constraint is binding. Let (p_1, p_2) denote the world market price path at the optimal mechanism and $(\hat{p}_1, \hat{p}_2)$ the world market price path in the absence of any mechanism. Then we have $p_2 > \hat{p}_2$ and $p_1 > \hat{p}_1$.*

Proof. See Appendix A.16 $\square$

Lemma 15 (The Monotone Mitigation Lemma, version with commitment and binding no-borrowing constraint). *Suppose that $T = 2$. Suppose that the global institution can fully commit but cannot borrow. Suppose that the no-borrowing constraint is binding in period 1. Let $X_t(F_1, F_2)$ denote aggregate cumulative coal extraction after period t given that the global institution’s funding is F_1 in period 1 and F_2 in period 2.*

We have :

$$X_1(F_1, F_2) = X_1(F_1, 0) \forall F_1, F_2$$

¹¹This argument also suggest that the Time Inconsistency Corollary 6 could be strengthened to a claim that if the global institution reoptimizes from a period $t > 1$ onwards (instead of just for period T as in the statement of the corollary) then it will spend less on rewarding supply reduction than under the initially announced mechanism.

$X_1(F_1, F_2)$ is strictly decreasing in F_1
 $X_2(F_1, F_2)$ is strictly decreasing in F_1 and F_2
 $X_2(F_1, F_2) - X_1(F_1, F_2)$ is strictly decreasing in F_2

Corollary 7 (The Time Inconsistency Corollary, under binding no-borrowing constraint). *Suppose $T = 2$ and suppose that the global institution cannot borrow in period 1 against the future income from period 2. Suppose at the optimal mechanism with full commitment the “no-borrowing constraint” is binding. Then if the global institution reneges on its promised mechanism in period 2 and reoptimizes it will spend less on rewarding supply reduction than the originally announced mechanism.*

Proof. By the Price Rise Lemma 14, the second period price under the optimal mechanism with full commitment would be higher than the second period price in the absence of any mechanism. But after period 1 the cumulative coal extraction will be strictly lower than in the absence of any mechanism, by the Monotone Mitigation Lemma 5.2. Thus by the Price Preservation Lemma applied to period 2, the price under the reoptimized mechanism will be lower than in the absence of any mechanism. $\square$

5.2 The optimal mechanism when the global institution cannot commit and cannot borrow

We have seen that if the global institution announces the optimal mechanism under full commitment then it subsequently has an incentive to renege on it and shift to spending less on rewarding supply reduction than initially announced (see corollary 6 for the case without borrowing/saving constraints and 7 for the case with binding no-borrowing constraints). This means that the global institution cannot implement the optimal mechanism if it cannot commit.

The question arises as to how large the welfare losses are from an inability to commit. This section tries to make progress towards answering this question.

Lemma 16 (The Surjectivity Lemma, dynamic version with no commitment and no borrowing or saving). *Suppose that the global institution cannot commit and cannot borrow. Consider a period t . Given the vector $(x_{it-1})_{i \in I}$ of the countries’ cumulative coal extraction until then, the global institution’s problem in period t is equivalent to choosing world market price p_t and $(x_{it}, y_{it}, z_{it})_{i \in I}$.*

Consider a combination of $(x_{it}, y_{it}, z_{it})_{i \in I}$ satisfying $\sum_{i \in I} x_{it} - x_{it-1} - y_{it} + z_{it} = 0$ and world market price p_t . Then for any set reward payment schemes implementing this combination in period t , the global institution must end up paying at least the amount $F_{iyt} := \sup_y b_{it}(y) - p_t y_{it} - (b_{it}(y_{it}) - p_t y_{it})$ for demand side contracting with country i in period t and at least the amounts $F_{izt} := \sup_z p_t z - G_i(z) - (p_t y - G_i(y))$ for substitute contracting with country i in period t .

Moreover, the global institution must at least spend the following amount F_{ixt} on rewarding country i in period t for supply reduction, where F_{ixt} is defined as follows:

$$F_{ixt} := (p_t - \frac{1}{1+r} p_{t+1})(x_{it}^*(p) - x_{it}) + (c_{it}(x_{it}) - c_{it}(x_{it}^*(p))) + \frac{1}{1+r} (c_{it+1}(x_{it}^*(p)) - c_{it+1}(x_{it})) \text{ for } t \in \{1, \dots, T-1\}$$

$$F_{ixT} := p_T(x_{iT}^*(p) - x_{iT}) + (c_{iT}(x_{iT}) - c_{iT}(x_{iT}^*(p)))$$

Furthermore, there does indeed exist a set of reward payment schemes for period t implementing $(x_{it}, y_{it}, z_{it})_{i \in I}$ and the price p_t that ends up paying exactly the amounts F_{iyt} on demand side contracting with i in t , the amounts F_{izt} on substitute contracting with i in t and the discounted amount F_{ixt} with i on spending on rewarding supply reduction in period t . Furthermore, there exists a set of positive affine linear reward payment schemes achieving this.

Proof. See Appendix A.17. $\square$

Lemma 17 (The Monotone Mitigation Lemma, version with no commitment and no borrowing or saving). Suppose that $T = 2$. Suppose that the global institution cannot commit and cannot borrow or save. Suppose that the global institution acts optimally under these constraints. Let $X_t(F_1, F_2)$ denote aggregate cumulative coal extraction after period t .

We have :

$$X_1(F_1, F_2) = X_1(F_1, 0) \forall F_1, F_2$$

$$X_1(F_1, F_2) \text{ is strictly decreasing in } F_1$$

$$X_2(F_1, F_2) \text{ is strictly decreasing in } F_1 \text{ and } F_2$$

$$X_2(F_1, F_2) - X_1(F_1, F_2) \text{ is strictly decreasing in } F_2$$

Proof. See Appendix A.18. $\square$

Lemma 18 (The Price Drop Lemma, version with no commitment and no borrowing or saving). : Suppose that the global institution cannot commit and cannot borrow or save. Suppose that the global institution acts optimally under these constraints. Suppose $T = 2$. Let $p_t(F_1, F_2)$ denote the price of energy in period t .

$$p_1(F_1, F_2) = p_1(F_1, 0) \forall F_1, F_2$$

$$p_2(F_1, F_2) = p_2(F_1, 0) \forall F_1, F_2$$

$$p_1(F_1, F_2) \text{ is strictly decreasing in } F_1$$

$$p_2(F_1, F_2) \text{ is strictly decreasing in } F_1$$

Proof. See Appendix A.18. $\square$

Let us restate a part of the Monotone Mitigation Lemma in terminology common in the literature:

Corollary 8 (The No-Green-Paradox Corollary for $T=2$). Suppose that the global institution cannot commit and cannot borrow or save. Suppose that the global institution acts optimally given these constraints. Suppose that $T = 2$. Then the weak Green Paradox cannot occur in the sense that increasing the budget at period 1 or period 2 cannot strictly increase coal extraction in period 1.

Corollary 9 (*The Endogenous Weak Green Paradox Corollary for $T=3$*). Suppose that the global institution cannot commit and cannot borrow or save. Suppose that the global institution acts optimally given these constraints. Suppose that $T = 3$. Then we have:

$X_1(F_1, F_2, F_3)$ is strictly increasing in F_2 .

Proof. See Appendix A.19. $\square$

The following proposition is reassuring and utterly unsurprising:

Proposition 1 (*The Monotone Eventual Emissions Proposition for $T=3$*). Suppose that the global institution cannot commit and cannot borrow or save. Suppose that the global institution acts optimally given these constraints. Suppose that $T = 3$. Let $x_t := \sum_{i \in I} x_{it}$. We have:

$\frac{\partial x_3}{\partial F_t} |_{(F_1, F_2, F_3)} < 0 \forall t \in \{1, 2, 3\}$ for any (F_1, F_2, F_3) such that the global institution's budget constraint is binding in each period.

Proof. See Appendix A.20 $\square$

We will now study under what condition increasing funding at some period would increase climate change damages. In line with the common usage of the term “Strong Green Paradox” (van der Ploeg and Withagen (2015)) we define:

Definition 7. We say that “the Strong Green Paradox occurs” if $\frac{d\eta}{dF_1} > 0$ or $\frac{d\eta}{dF_2} > 0$ or $\frac{d\eta}{dF_3} > 0$. Moreover, we say that “the Strong Green Paradox occurs for small budgets” if $\frac{d\eta}{dF_1} |_{(F_1, F_2, F_3)=0} > 0$ or $\frac{d\eta}{dF_2} |_{(F_1, F_2, F_3)=0} > 0$ or $\frac{d\eta}{dF_3} |_{(F_1, F_2, F_3)=0} > 0$.

For tractability, we will focus on the case where F_1, F_2, F_3 are all small.

Lemma 19. Suppose $T = 3$. Suppose the global institution cannot commit at all. We always have $\frac{d\eta}{dF_1} |_{(F_1, F_2, F_3)=0} \leq 0$ and $\frac{d\eta}{dF_3} |_{(F_1, F_2, F_3)=0} \leq 0$. In particular, the Strong Green Paradox occurs for small budgets iff $\frac{d\eta}{dF_2} |_{(F_1, F_2, F_3)=0} > 0$.

Proof. See Appendix A.21 $\square$

The sign of $\frac{d\eta}{dF_2}$ is determined by two effects. Firstly, any increase in funding for the global institution will decrease the amount of eventual emissions, by the Monotone Eventual Emissions Proposition 1. This is good, since by assumption 5 $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$ is increasing in its last argument. On the other hand, by the Endogenous Weak Green Paradox Corollary 9 we know that $\frac{dx_1}{dF_2} > 0$. This is bad, since by assumption 5, $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$ is increasing not just in its last argument but also in all of its other arguments.

To determine which of these effects dominates, we need to make further assumptions on $\eta(\sum_{i \in I} x_{i1}, \dots, \sum_{i \in I} x_{iT})$. In other words: we need to make an assumption about how valuable it is to delay global warming. For concreteness

and ease of exposition, I will from now on assume that delaying global warming gives returns corresponding to the interest rate¹²:

Assumption 7. $\eta(x_1, x_2, x_3) = \tilde{\eta}(x_1 + \frac{1}{1+r}(x_2 - x_1) + \frac{1}{(1+r)^2}(x_3 - x_2))$, where $\tilde{\eta}$ is strictly increasing.

Proposition 2 (Necessary and Sufficient Conditions for the Strong Green Paradox to not arise for Small Budgets). Let us denote by $e_{xt} := \frac{dx_t^*}{dp_t} \frac{p_t}{x_t^*}$ and $e_{yt} := -\frac{dy_t^*}{dp_t} \frac{p_t}{y_t^*}$. Suppose that assumption 7 holds. Suppose the global institution cannot commit at all. Define the following constant a :

$$a := \frac{e_{x1}e_{x2}e_{y1}p_3x_1x_2}{(x_3-x_2)((e_{x1}+e_{y1})(e_{x2}x_2+e_{y2}(x_2-x_1))p_1(1+r)+e_{x1}e_{y1}x_1p_2)}$$

or equivalently:

$$a := e_{y1} \frac{1}{\left(\frac{x_3-x_2}{x_1}\right)\left(1+\frac{e_{y1}}{e_{x1}}\right)\left(1+\frac{e_{y2}}{e_{x2}}\left(\frac{x_2-x_1}{x_2}\right)\right)\frac{p_1(1+r)}{p_3}+\frac{e_{y1}}{e_{x2}}\frac{x_1}{x_2}\frac{p_2}{p_3}}$$

Then the following condition is sufficient for the Strong Green Paradox to not occur for small budgets:

$$e_{y3} \geq a$$

Moreover, if $e_{y3} < a$ then the following condition is necessary and sufficient for the Strong Green Paradox to not occur for small budgets:

$$e_{x3} \geq \frac{\frac{a-e_{y3}}{a} e_{x1}e_{x2}e_{y1}r x_1x_2p_3}{x_3((e_{x1}+e_{y1})((x_2-x_1)e_{x2}+x_2e_{y2})p_1(1+r)^2+e_{x1}e_{y1}x_1p_2(1+r))}$$

or equivalently:

$$e_{x3} \geq e_{y1} \frac{\frac{a-e_{y3}}{a} r}{\frac{x_3}{x_2}\left(1+\frac{e_{y1}}{e_{x1}}\right)\left(\frac{x_2-x_1}{x_1}+\frac{x_2}{x_1}\frac{e_{y2}}{e_{x2}}\right)\frac{p_1(1+r)^2}{p_3}+\frac{e_{y1}}{e_{x2}}\frac{p_2(1+r)}{p_3}}$$

Proof. I computed all this with Mathematica, building on the computations explained in the proof of proposition 1. The documented Mathematica notebook is available for download here □

¹²This assumption stacks the odds in favor of a Strong Green Paradox arising, as I will now argue. One interpretation of this assumption is that climate change damages at a given time is directly proportional to cumulative emissions at that time and that these damages should be discounted using the market interest rate. Both of the linearity assumption and the discounting assumption will make it more likely for the Strong Green Paradox to occur than under plausible alternative assumptions. I will now discuss each of these two assumptions.

The translation of cumulative emissions into temperature is commonly assumed to be concave whilst the translation of temperature into damages is widely assumed to be convex. Concerning the former, the IPCC has recently concluded: “Multiple lines of evidence indicate a strong, consistent, almost linear relationship between cumulative CO2 emissions and projected global temperature change to the year 2100 .” IPCC (2014a, p. 9). Since damages are plausibly strongly convex in temperature (Bretschger and Pattakou (2018)), this implies that damage is a convex function in cumulative emissions.

By assuming away this convexity, I make it more likely for the Strong Green Paradox to arise. To see why, we note that since in the short term global warming will be less severe, the linear specification will overestimate the short term effects relative to the longer term effects. But the Strong Green Paradox arises if the harm due to increased global warming in the short term outweighs the benefit from reducing global warming in the medium and long term. It is therefore more likely to occur under the linear specification than under a more realistic convex specification for the dependence of damages on temperature.

Now let us turn to the discounting assumption. Normatively, we should discount future monetary damages only due to considerations about decreasing marginal utility and due to extinction risk. As a result, we should use a discount rate lower than the market interest rate. Doing so would again lead us to weigh more heavily longer term climate damages relative to short term damages. Thus the Strong Green Paradox would also be less likely to occur.

Corollary 10. *The following is a sufficient condition for the Strong Green Paradox to not arise:*

$$(c_1''(x_1) - \frac{1}{1+r}c_2''(x_1))(\frac{x_2-x_1}{x_2})\frac{(1+r)^2}{r} + (c_2''(x_2) - \frac{1}{1+r}c_3''(x_2))\frac{1+r}{r} \geq c_3''(x_3)$$

Proof. See appendix . □

Corollary 11. *Suppose that extraction costs do not change over time, so that we can denote them simply by $c(x)$. The the following is a sufficient condition for the Strong Green Paradox to not arise for small budgets:*

$$c''(x_1)(\frac{x_2-x_1}{x_2})(1+r) + c''(x_2) \geq c''(x_3)$$

Proof. Follows directly from previous corollary. □

It is important to highlight here that the sufficient condition provided in the preceding corollary is in general far from being necessary. In fact the condition is only necessary if one makes the worst possible (from the perspective of someone hoping that the Strong Green Paradox will not arise) assumptions on the demand elasticities (namely $e_{y1} = \infty, e_{y2} = 0, e_{y3} = 0$). However, remarkably, the condition allows us to conclude that if marginal cost is an affine linear function of cumulative emissions then the Strong Green Paradox cannot occur for small budgets (since in that case in particular we would have $(c''(x_1) = c''(x_2) = c''(x_3))$).

However, there is evidence that the true long term global marginal cost curve for coal extraction might actually be convex. For the rest of this section I will assume that the marginal cost curve for coal is given by the diagram below (taken from Wood Mackenzie (2016)):

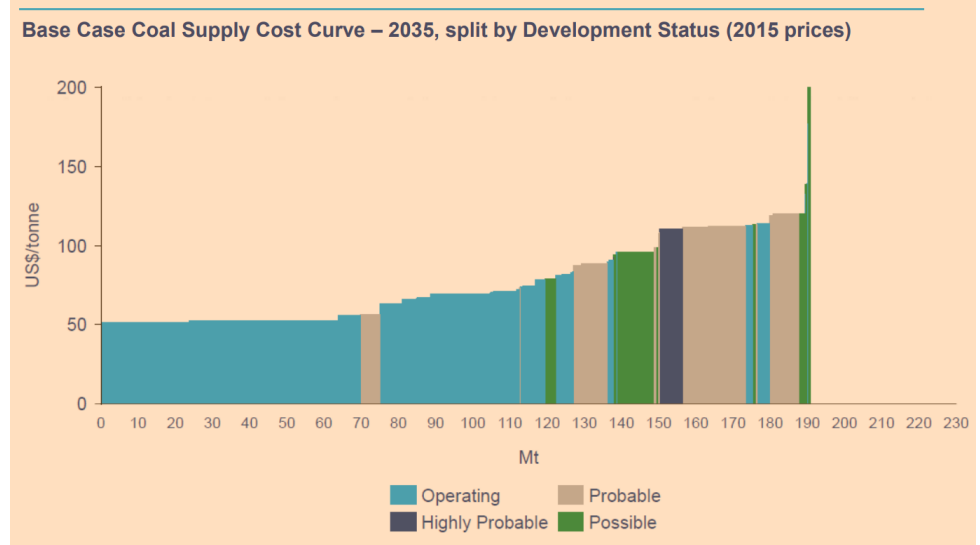


Figure 3:

In order to apply the next proposition, we will need to have a cost function $c(x)$ that is twice differentiable. For that purpose, I have taken five points (at 0,50,100,150 and 190) from the above estimated coal supply curve and computed the cubic that best fits them (in the sense of least squares) under the constraint of having positive slope on the entire domain of the cost function (the interval $[0, 190]$). The result is the following:

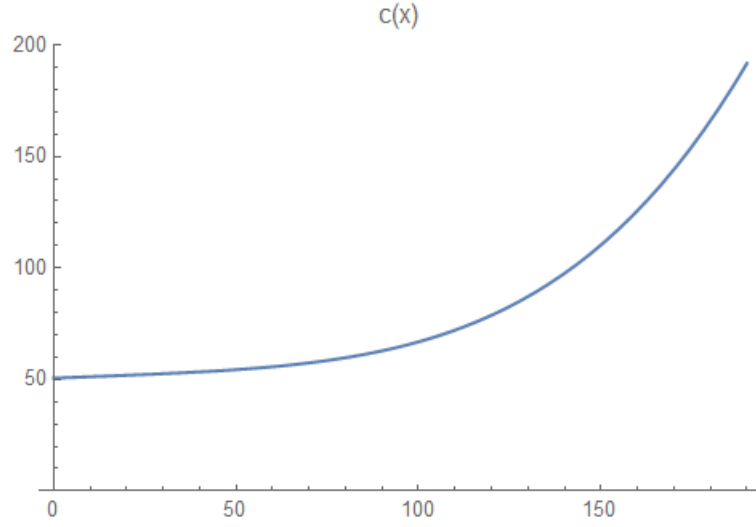


Figure 4:

Proposition 3. *Suppose that extraction costs do not change over time, so that we can denote them simply by $c(x)$. Then the Strong Green Paradox occurs for small budgets in the three period model iff $h := g_1 g_2 + g_3 + g_4 (g_5 + g_6) < 0$, with the following definitions:*

$$g_1 := r c''(x_2) (e_{y3} r (x_3 - x_2) c''(x_3) + (1 + r) c'(x_3))$$

$$g_2 := (e_{y1} x_1 + e_{y2} (x_2 - x_1)) (r c'(x_2) + c'(x_3)) - e_{y2} r (1 + r) (x_1 - x_2) c'(x_1)$$

$$g_3 := e_{y1} r (1 + r) x_1 c''(x_1) (e_{y3} r (x_3 - x_2) c''(x_3) + (r + 1) c'(x_3)) (e_{y2} r (x_2 - x_1) c''(x_2) + r c'(x_2) + c'(x_3))$$

$$g_4 := r c'(x_2) + c'(x_3)$$

$$g_5 := (1 + r) c'(x_3) (r (1 + r) c'(x_1) + r c'(x_2) + c'(x_3))$$

$$g_6 := rc''(x_3)(e_{y3}(x_3 - x_2)(r(1+r)c'(x_1) + rc'(x_2) + c'(x_3)) - e_{y1}(1+r)x_1c'(x_3))$$

Proof. Provided in the Mathematica notebook downloadable [here](#). $\square$

Proposition 3 tells us precisely whether the Strong Green Paradox will occur for small budgets. Given an assumed cumulative cost function $c(x)$, all that we have to input are the demand elasticities e_{y1}, e_{y2}, e_{y3} , the cumulative coal extraction sequence x_1, x_2, x_3 and the interest rate r . The quantities e_{y1}, e_{y2}, e_{y3} and x_1, x_2, x_3 are of course all endogenous quantities. They will depend on the entire demand functions in the three periods. One might therefore be inclined to think that proposition 2 is not very useful for assessing whether the Strong Green Paradox can plausibly occur or not. The pessimistic reader might be impatient to find a result that gives a condition as a function of the primitives of the model instead, i.e. as a function of the demand functions.

However, I will now argue that proposition 3 actually allows us to conclude that the Strong Green Paradox is extremely unlikely to occur, at least for the extraction cost function $c(x)$ shown above.

Corollary 12. *Suppose that the cost function $c(x)$ is as shown in figure 5.2. Moreover, suppose that the third period extraction is not more than the second period extraction. Then the Strong Green Paradox can only occur for small budgets if $\frac{e_{y1}}{e_{y2}} > 15.55$ or $\frac{e_{y1}}{e_{y3}} > 15.55$.*

Proof. Provided in the Mathematica notebook downloadable [here](#). $\square$

The assumption stipulated in corollary 12 that the third period extraction is not more than the second period extraction is plausible: The three periods are of equal duration (since the interest rate r is the same) and the third period is by assumption the last when coal is extracted.

How plausible is it that $\frac{e_{y1}}{e_{y2}} > 15.55$ or $\frac{e_{y1}}{e_{y3}} > 15.55$? The price elasticities of demand e_{y1}, e_{y2} and e_{y3} are determined by how costly it is for consumers and firms to switch to alternatives to coal or to consume less energy. This might change over time, but we should not expect conditions to change so much that e_{y2} or e_{y3} would differ by a factor of 15.55 from e_{y1} ! Thus we can conclude that the Strong Green Paradox cannot arise under any plausible assumptions on the demand side, at least if we assume that the cost function $c(x)$ is given as in Figure 5.2.

6 Numerical illustrations for the dynamic model under constant elasticity specification for renewable energy and for energy demand

I have not yet identified the appropriate projections for the relevant demand and supply curves that are required for applying the dynamic model. I now

provide for illustration a numerical example of the model with three periods. The specific assumptions used are documented in the Mathematica notebook that can be downloaded [here](#). For the curve determining the marginal extraction cost of coal I assumed the one given in figure 5.2. The endogenous extraction path is as follows:

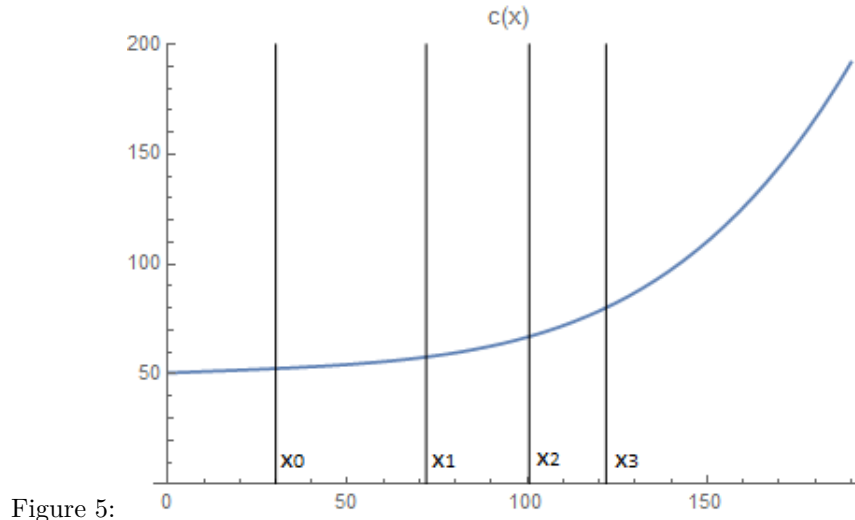


Figure 5:

In the Mathematica notebook, I consider the case where the global institution fully commits and can freely save and borrow. I compute the paths of spending by the global institution at the optimal mechanism. For the numerical example I take the case where the global institution's (intertemporal) budget is small. The results are shown here:

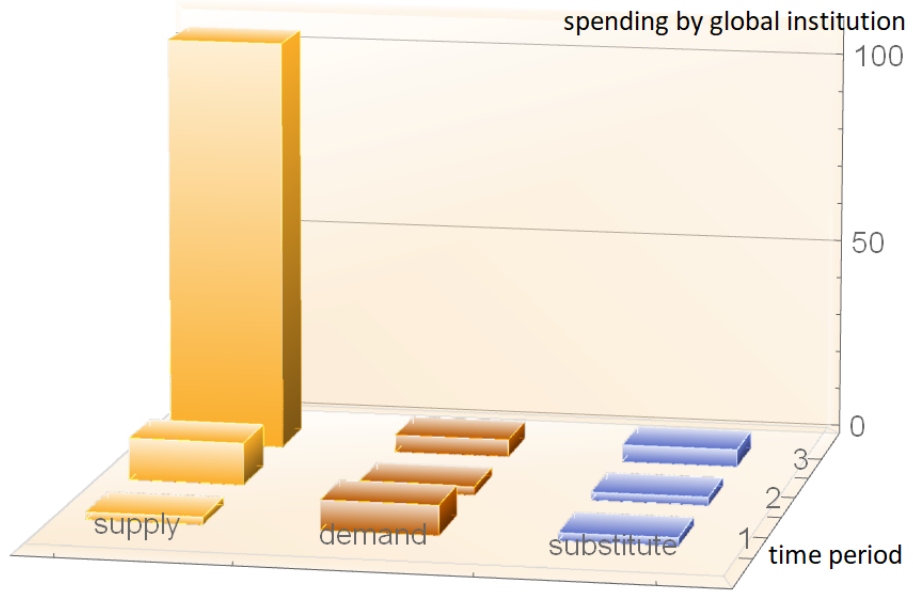


Figure 6:

This numerical example hints at the significance of the commitment problem that we identified in section 5: The optimal mechanism with full commitment involves spending most of the available intertemporal budget on rewarding supply reduction in the last period. However, by corollary 6, we know that once the last period arrives, the global institution always has an incentive to renege on its promised reward payment schemes and to shift money away from spending on rewarding supply reduction.

7 Endogenous Funding

So far, the entire analysis has assumed that funding for the global institution is exogenous. Now I will endogenize this. I will compare endogenous funding under two different architectures:

architecture 1: *There is a separate global institution for rewarding countries for supply reduction, demand reduction and substitute expansion. We refer to these institutions as “the supply reduction reward fund”, “the demand reduction reward fund” and “the substitute expansion reward fund”. (In the financing game analyzed below, the budget split between demand reduction and substitute expansion will at equilibrium always be optimal. Thus we can in the context of the game treat them as a single institution and simply refer to it as “the demand reduction reward fund”.) Donors can earmark their contributions to the specific institutions.*

architecture 2: *There is a single global institution that uses its available*

budget in the optimal way for rewarding countries for supply reduction, demand reduction and substitute expansion. Donors cannot earmark their contributions.

7.1 The financing game

7.1.1 Payoffs

The countries of the world are exogenously partitioned into a set J of players. There are two groups of players, the net coal exporters and the net coal importers: $J = J_{\text{net coal exporters}} \cup J_{\text{net coal importers}}$. The intended interpretation here is: a player is comprised of countries who decide individually on their respective coal extraction, energy use and renewable energy production taking the reward payment schemes as given but (in line with the preceding sections) but who act jointly as unitary players when it comes to deciding on how much money to give to the funds for rewarding supply reduction and demand reduction. The motivation for this interpretation is that it might be easier to cooperate on monetary contributions (which are directly comparable and that can be flexibly adjusted) than on individual domestic actions about fossil fuel extraction and energy use (which are harder to compare and that can only be adjusted with substantial time lags).

Player j chooses the monetary contribution t_{xj} that it makes to the supply reduction reward fund and the monetary contribution t_{yj} that it makes to the demand reduction reward fund. Thus the strategy profile in this financing game is $t := (t_{xj}, t_{yj})_{j \in J}$. The players anticipate how the aggregate funding for the two funds, $t_x := \sum_{j \in J} t_{xj}$ and $t_y := \sum_{j \in J} t_{yj}$ determine the world market price of coal, $p(t_x, t_y)$ and the individual countries' choices, $y_i(t_x, t_y)$. Player j 's payoff in this financing game is thus:

$$U_j(t) = -t_{xj} - t_{yj} + B_j(y_j(t_x, t_y)) - C_j(x_j(t_x, t_y)) + p(t_x, t_y)(x_j(t_x, t_y) - y_j(t_x, t_y)) + f_j(x_j(t_x, t_y), y_j(t_x, t_y), t_x, t_y) + \eta_j x_j(t_x, t_y)$$

7.1.2 Timing: randomly alternating adjustment moves

Time runs discretely: $\tau = 1, 2, \dots$. We denote the time period with superscripts. In period 1 players play some initial profile t^1 . In each subsequent period τ , a random player j is selected and given the opportunity to adjust his strategy. All the other players continue playing the strategy that they played in the previous period. Player j discounts the future using the interest rate r , giving the intertemporal payoff $\sum_{\tau=1}^{\infty} \frac{1}{(1+r)^\tau} U_j^\tau$.

We will assume that in each period the game is given by the static model that we analyzed in section 3. We thus abstract away from the dynamic issues analyzed in sections 5 and 4.

7.1.3 Solution concept: Backward Induction Induced Markov Perfect Equilibrium (BIIMPE)

At any stage of the game, the payoff relevant state is given by ("player who's turn it is to adjust his action", "action profile of the other players"). A Markovian

strategy for a player is thus a mapping specifying the action (or probabilistic mixture of actions) to which to adjust his strategy if given the chance to do so and depending on the profile of the other players' strategies.

Definition 8. Consider the following auxiliary truncated version of the dynamic game: time ends at 0 and starts at $\underline{\tau} < 0$. A Backward Induction Induced Markov Perfect Equilibrium (**BIIMPE**) is a profile of Markovian strategies that is the limit of some backward induction solution of the truncated game as the starting period $\underline{\tau}$ goes to $-\infty$.

7.2 A specification

For the numerical computations shown below, we use the following specification

$$U_j(x_i, y_i, p) = \phi_i B\left(\frac{y_i}{\phi_i}\right) - \psi_i C\left(\frac{x_i}{\psi_i}\right) + p(x_i - y_i) + f_i(x_i, y_i) + \eta_j \sum_{j \in J} x_j$$

with $\sum_{j \in J} \phi_j = \sum_{j \in J} \psi_j = 1$ and $\sum_{j \in J} \eta_j = \eta$.

Thus ϕ_j is the proportion of the coal combustion happening in j , ψ_j is the proportion of coal extraction happening in j and $\frac{\eta_j}{\eta}$ is the proportion of climate change damages accruing to j .

For ease of interpretation, I now make the following symmetry assumptions:

$$\begin{aligned} \sum_{j \in J_{\text{net coal exporters}}} \phi_j + \psi_j &= \sum_{j \in J_{\text{net coal importers}}} \phi_j + \psi_j \\ \sum_{j \in J_{\text{net coal exporters}}} \eta_j &= \sum_{j \in J_{\text{net coal importers}}} \eta_j \\ \frac{\phi_j}{\psi_j} &= \frac{\phi_k}{\psi_k} \text{ for } j, k \in J_{\text{net coal exporters}} \text{ or } j, k \in J_{\text{net coal importers}} \\ \frac{\eta_j}{\psi_j} &= \frac{\eta_k}{\psi_k} \text{ for } j, k \in J_{\text{net coal exporters}} \text{ or } j, k \in J_{\text{net coal importers}} \end{aligned}$$

Now let us impose that in the absence of any global institution rewarding countries for supply reduction or substitute expansion, global coal exports make up 20% of global coal production, as is in fact the case as of 2020 (IAEA 2020). Then we get:

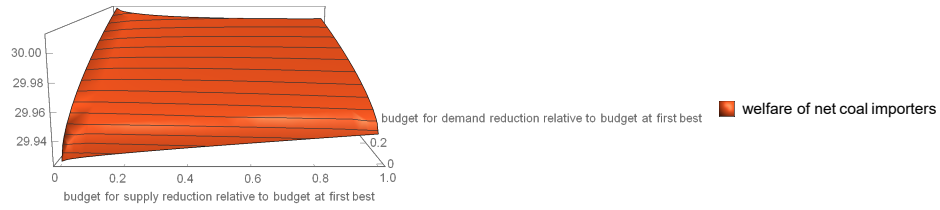
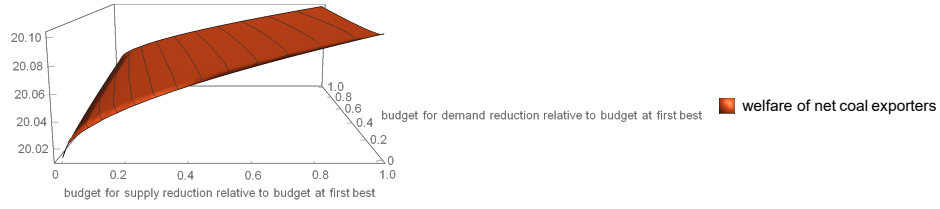
$$\frac{\phi_j}{\psi_j} = \frac{2}{3} \text{ for } j, k \in J_{\text{net coal exporters}} \text{ and } \frac{\phi_j}{\psi_j} = \frac{3}{2} \text{ for } j, k \in J_{\text{net coal importers}}$$

Given the above assumptions, we can view $\frac{\eta_j}{\eta}$ as the “size of player j ”.

Lemma 20. Consider a hypothetical one shot game where payoffs are as given in section 7.1.1 and all players simultaneously decide on their actions (t_{xj}, t_{yj}) . Then at any Nash equilibrium we have: All players that are not the players of largest size in their group (net coal exporters or net coal importers) choose $(0, 0)$.

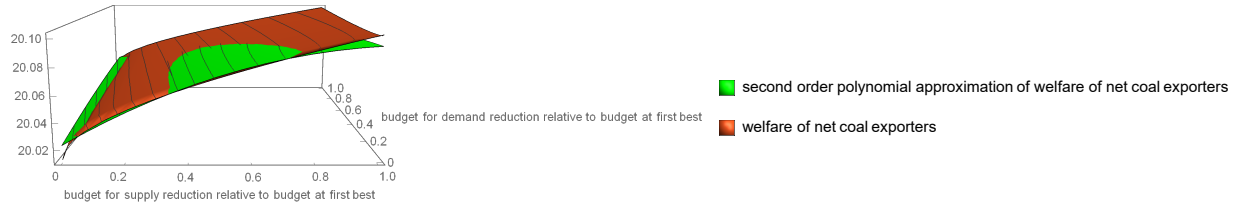
7.2.1 Numerical Payoffs

For the numerical computation of the payoffs I assume that the price elasticity of coal demand is 1.02, based on Teng et al. (2019) and that the price elasticity of coal supply is 1.3 based on Dahl (2009). The surfaces shown in red below give the welfare accruing to net coal exporters as a function of the budget t_x of the fund rewarding supply reduction and the budget t_y of fund rewarding demand reduction (not taking into account the fact that the budgets must be provided by the players).



These plots show that for the net coal exporters welfare increases much more in the budget for the supply-reduction reward fund than in the budget for the demand-reduction reward fund. The converse holds for the net coal importers. This is because the former increases the price of coal whereas the latter decreases it.

In the graph below I show in green the second order polynomial in t_x and t_y that best approximates the welfare of net coal exporters (in a least squares sense) on the domain given by $t_x \geq 0, t_y \geq 0$ and the condition that $t_x + t_y$ does not exceed the aggregate budget required to implement the first best.



Similarly, here is the second order polynomial best approximating the welfare of net coal importers, shown in green:



The second order polynomials are excellent approximations, as can be seen in more detail in the Mathematica notebook provided here. I will from now

on use the second order polynomials instead of the true functions that they approximate. This will allow me to exactly compute the BIIMPEs.

7.3 BIIMPE results

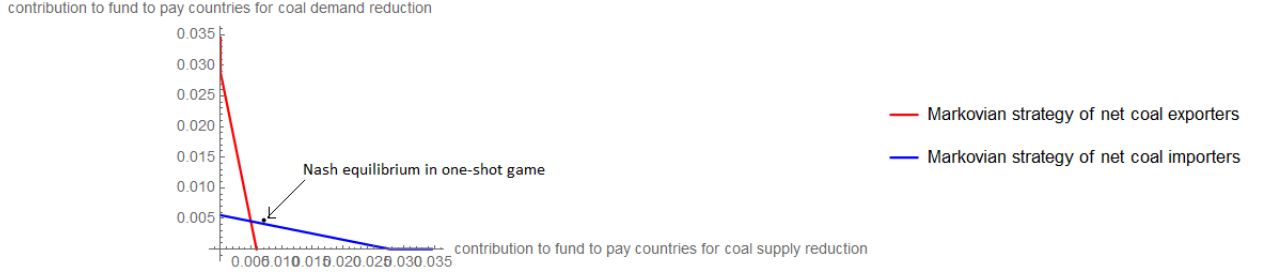
From now on I assume that only the largest net coal exporter and the largest net coal importer participate in the financing game and that they have the same size. I conjecture that the results that I will present below would be the same in the full game where all players can contribute money.

7.3.1 The case of separate funds for rewarding countries to reduce supply and demand (architecture 1)

In the Mathematica notebook downloadable here I computed the backward induction solution of the truncated game. I assumed that the discount factor is $\beta = 0.95$, which would correspond to assuming that the players use a discount rate of 5% and adjust their contributions on average every two years. The results hardly change for $\beta \in [0.8, 1]$.

When the size of the largest players is at least 25 % of the world then strategies at each stage of the backward induction are piecewise linear like in the plot below. They converge to a profile of Markovian strategies which form thus the unique BIIMPE. This unique BIIMPE for the case where the largest players each comprise exactly 25% of the world is shown here:

Figure 7: BIIMPE for when each of the two largest players comprise 25% of the world



The red line records as the x-coordinate how much the net coal exporter gives to the supply reduction reward fund, given that it is his turn and given that the net coal importer is giving the amount corresponding to the y-coordinate to the demand reduction reward fund. The Markov chain induced by this BIIMPE converges to the unique point of intersection of the red and the blue lines. This point is close to the point corresponding to the Nash equilibrium in the (hypothetical) one shot simultaneous move game.

The BIIMPE corresponds to all players being fully rational. Each player takes into account that by giving more to his preferred fund, he will affect the other player's incentives to give to the latter's preferred fund in the future.

This is not taken into account by players at the static Nash equilibrium in the hypothetical one shot simultaneous move game. There are decreasing marginal returns from money spent via the funds, as is reflected in the concavity of the welfare surfaces shown in section 7.2.1. This suggests a consideration against giving much: By giving more, a player will decrease the incentives for the other player to give. Since this consideration is absent in the static Nash equilibrium logic, it is therefore surprising that the Markovian strategies at the BIIMPE shown above lead to a point almost as high as the static Nash equilibrium.

The explanation for this is that there is a countervailing effect: If the coal importers give more to the demand reduction reward fund then the world market price for coal declines. Thus the coal exporters are less concerned about crowding out the coal importers' contributions than they would be if there was a single combined fund with a neutral effect on world market prices (as would be the case for a single combined fund maximizing global welfare for its given budget, by the Price Preservation Lemma 3).

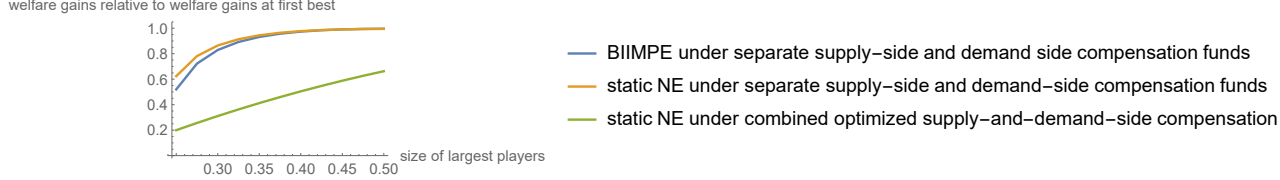
7.3.2 The case of a single combined optimized supply-reduction-and-demand-reduction reward fund fund (architecture 2)

In this case it turns out that the backward induction solution of the truncated game does not converge as the starting time, τ , goes to $-\infty$. Thus in particular, there is no BIIMPE and this concept cannot be directly used to provide a prediction of the eventual outcomes. However, it turns out that all along the backward induction solution the aggregate contributions to the combined fund will never exceed the aggregate contributions that arise at the static Nash equilibrium in the hypothetical one shot game where both players decide on their contributions simultaneously. In appendix A.23 I illustrate this result. I also prove for any Markov Perfect Equilibrium that if the induced Markov Chain converges to a point then at that point the aggregate contributions must be strictly lower than at the static Nash equilibrium (see Lemma 22).

7.4 Welfare comparison between architecture 1 (“separate supply-reduction reward fund and demand-reduction reward fund”) and architecture 2 (“combined optimized supply-reduction-and-demand-reduction reward fund”)

The following graph shows the welfare gains realized relative to the situation without any reward funds under different scenarios. They are normalized so that the welfare gains achieved at the first best are set to 1. The blue and orange lines summarize results obtained for architecture 1. Shown in blue is the outcome at the profile of contributions that the Markov chain induced by the unique BIIMPE would converge to. Shown in orange is what would result if at each stage of the game all players would play according to the unique Nash equilibrium in the hypothetical one shot simultaneous move game.

The green line summarizes results obtained for architecture 2. Lemma 21 implies that backward induction reasoning analogous to that of the BIIMPE could only lead to even worse results.



These results suggest that architecture 1 greatly outperforms architecture 2. This is despite the fact that under architecture 2 the money is split between the demand reduction and supply reduction funds in a way that maximizes global welfare, whereas this is not the case under architecture 1. The reason is that under architecture 1 much more money is raised, which outweighs the inefficiency from suboptimal allocation between the demand reduction and supply reduction funds.

8 Implications for the design of global public good institutions and mechanisms to fund them

8.1 Comparing architecture 1 and 2 if the global institutions are financed via voluntary contributions

The primary way that global environmental institutions (and Global Public Good Institutions generally) are currently being financed is through assessed contributions and voluntary contributions by governments. Many global institutions have a wide range of different programs. For example, the Green Climate Fund finances both projects that can be viewed as rewarding countries for fossil fuel demand reduction and projects that can be viewed as rewarding countries for fossil fuel substitute expansion. An important design question is whether donors should be allowed to earmark their contributions to specific programs.

In the case of institutions for Global Health, countries have been enabled to earmark their contributions¹³. However, in the case of the Green Climate Fund donors can earmark only 20% of contributions.

A consideration weighing in favor of allowing countries to earmark their contributions is that this could increase the amounts that they will be willing to donate.¹⁴ Statistical evidence suggests that this is true for international organizations generally (Bayram and Graham (2017)). One explanation for this

¹³see Clinton and Sridhar (2019) for a critical take on this.

¹⁴For example, Raman (2014) reports on the negotiations about rules for earmarking for the Green Climate Fund: “Several interested contributors from developed countries led by the UK, US and Norway expressed the view that some targeting of funds should be allowed as this would enable more resources to come into the Fund.”

could be that donors have different beliefs about the effectiveness of different programs. A further explanation, highlighted in section 7, is that different approaches to curbing a good with global externalities will differently affect the world market price of this good. As a result, donors have self-interested reasons to prefer supply reduction approaches or demand reduction approaches, depending on whether they are net exporters or net importers of the good.

The results from section 4 suggest that it would be valuable to supplement the current global institutions that reward countries for demand fossil fuel demand reduction and substitute expansion with a global institution rewarding countries for supply reduction. Moreover, the results from section 7 suggest that it would be best for such a supply-side institution to be separate and independent so that countries can donate money specifically for their preferred approach. I have referred to this way of separating out the different institutions as “architecture 1”.

An alternative arrangement, which I have called “architecture 2” would be to create a unified global institution that would commit to splitting its budget between the different approaches to curbing fossil fuels in the way that is ex ante optimal. For example, the global fund could expand its scope to also reward countries for reducing fossil fuel supply and it could adopt a rule specifying how to split the budget, depending only on the aggregate funding flows that it receives. The Green Climate Fund already splits each year the funding flows that it receives between mitigation and adaptation projects in equal parts. It could adopt a further such rule for splitting the mitigation spending between supply reduction, demand reduction and substitute expansion. Moreover, this rule for splitting the funding could increase the proportion going to supply reduction over time, in line with the results from section 5.0.1 and 6.

The downside of this architecture 2 is that it would reduce donors’ incentive to contribute funding, given that they would not be able to earmark it. Section 7 suggests that this incentive effect is so strong that it would outweigh static efficiency gains that it would realize relative to architecture 1. However, the model from section 7 abstracted away (for tractability reasons) from the intertemporal nature of fossil fuel markets. It is therefore important to complement this analysis with an assessment of the size of the dynamic inefficiencies of architecture 1 that arise from the time consistency problem highlighted in section 5.2. Architecture 2, would by construction be immune to time consistency problems: Under this architecture, the rules for splitting the budget between the different approaches could be set in the ex ante optimal way.

Under architecture 1, on the other hand donors would plausibly not be able to commit to a path of funding to allocate to the different funds (rewarding supply reduction, demand reduction and substitute expansion). Donors with sufficiently large net fuel exports would always prefer giving to the fund rewarding countries for supply reduction, whilst donors with sufficiently large net fuel imports would always prefer giving to the fund rewarding countries for demand reduction. Donors primarily concerned with global welfare would plausibly want to commit ex ante to direct their flow of donations to the funds for demand reduction and substitute expansion initially but to switch to directing their flow

of donations to the fund for supply reduction later on. This is suggested by the illustrative numerical results shown in section 6 as well as Lemma 13. However, once the later periods arrive, these donors might have an incentive to renege on their announced plan. They might find it optimal to continue directing a part of their funding to the demand reduction fund and the substitute expansion fund instead of directing all their funding to the supply reduction fund as was part of their ex ante optimal plan.

This time consistency problem always arises if all donors contribute exogenous funding flows and are solely motivated by global welfare. I established this rigorously in the form of Lemmas 6 and 7. But it seems clear that this problem will still arise to some extent in the more realistic case where donors are partly motivated by promoting global welfare and their contributions are endogenous.

Given these observations, it would be useful to bound the extent of the welfare losses arising due to the time consistency problem. Corollary 12 suggests such a bound: It shows under the specified assumptions that additional funding at no period can lower global welfare under any plausible set of assumptions. This suggests that the efficiency losses due to the time consistency problem are unlikely to tilt the balance in favor of architecture 2, at least if the incentive benefits of architecture 1 over architecture 2 are as large as suggested by the results in section 7.

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8.2 Comparing architecture 1 and 2 if the global institutions are financed via other mechanisms

The discussion in the section 8.1 has compared architectures 1 and 2 (see the beginning of section 7 for definitions), assuming that the different reward funds are all only financed through voluntary contributions from governments. The model in section 7 aims to capture this form of financing, where countries revise

¹⁵The case for architecture 1 over architecture 2 would further be strengthened if it is possible to mitigate the time consistency problem through the design of additional institutions. One potential partial remedy would be the creation of a fund to buy up fossil fuel deposits. In the model from section 5, buying up coal deposits at time $t < T$ can be done so as to effectively add money to the supply side approach in period T . However, this cannot remedy the inefficiencies that come from the fact that in periods before T the global institution will (if it cannot commit) choose inefficiently low supply side budgets. Thus deposit purchase schemes can only be a partial remedy to the time consistency problem.

Another approach to mitigating the time consistency problem could be to establish so-called “Time-indexed Proportional Matching Funds” defined as follows: In year s countries could donate money to a Proportional Matching Fund (PMF) for year $t > s$ supporting the (fossil fuel) supply reduction reward fund and some other global institutions. The PMF would in year t split its budget between the institutions that it supports, in proportion to the voluntary contributions that these institutions receive in year t . This would in year t increase the incentives for the countries favoring the supply reduction reward fund to give to this institution, since by doing so the latter can cause a greater share of the money from the PMF to go to that institution. Thus by giving in year s to such a PMF indexed for time t countries can cause increases in funding for the supply reduction reward fund in period t .

I leave it for future research to systematically assess these and other approaches to remedy the time consistency problem under architecture 1.

every once in a while their flow of contributions, taking into account the other countries' current flows of contributions.

However, it might be possible in the future to establish new mechanisms to fund global public good institutions (GPGIs). Such mechanisms could be purposefully designed in such a way that participants could through their contributions affect the overall allocation of funding in specific ways so that participants would have strong incentives to participate and contribute. An example of a proposal for such a mechanism is the so-called “MGF mechanism with Proportional Matching Funds” proposed in Stern (2020), where I find the following: Under architecture 1 the mechanism can raise twice as much aggregate funding as under architecture 2¹⁶. This advantage of architecture 1 is likely to outweigh any loss in allocative efficiency, given the results from section 4: there we found that as long as each of the three GPGIs gets at least half its optimal share of funding, the global welfare losses relative to the optimum split of the available money is at most 12%. Moreover, taking into account the dynamic inefficiencies due to the time inconsistency problem studied in section 5 are unlikely to overturn this conclusion, given the reassuring results concerning the impossibility of a Strong Green Paradox under any plausible assumptions (see Corollary 1).

9 Conclusion and limitations

This paper has analyzed the problem faced by global institutions such as the Green Climate Fund, the Global Environment Facility and the Climate Investment Fund. With the part of the budgets allocated to climate change mitigation, these institutions can be viewed as being able to reward countries for reducing fossil fuel supply, reducing energy demand and expanding renewable energy. So far, these institutions only pursue the demand reduction and the substitute expansion approaches. The results in this paper suggest that it would be optimal for them to pursue a mixture of supply, demand and substitute based approaches to curbing fossil fuels.

Moreover, the results from the financing game presented here suggest that it would be best to enable countries to earmark their funding to a particular approach. This could be done by creating a separate institution for each approach. Net exporters of fossil fuels would then have a stronger incentive to give to the fund rewarding fossil fuel supply reduction, since this would increase fossil fuel world market prices. Net importers of fossil fuels would have a stronger incentive to give to the funds rewarding countries for fossil fuel demand reduction and substitute expansion, since this would decrease fossil fuel world market prices. An alternative centralized design, where all funding would be split between the different approaches so as to maximize global welfare would probably not raise as much funding. This would likely outweigh any efficiency gains it might achieve relative to a flexible decentralized architecture where countries

¹⁶To be more precise, in Stern (2020) I consider a more general version of architecture 1 and 2, applied to all GPGIs, not just the once trying to curb fossil fuels that are the focus of the current study.

can earmark their contributions to their preferred approaches.

An important limitation is that the model does not take into account the adverse effects of fossil fuel rents on global welfare documented in the literature on the natural resource curse (Ross (2015)). This consideration weighs in favor of focusing on demand reduction and substitute expansion instead of supply reduction. Possibly, it could rationalize the current absence of international institutions paying countries for reducing fossil fuel supply.

A further limitation is that the model abstracts away from informational asymmetries. Since in reality countries have private information about their costs and benefits of extracting coal, using energy and producing renewable energy, most of them will reap informational rents at the optimal mechanism. If the ratio of informational rents over the money used was the same across the three approaches (rewarding supply reduction, demand reduction, substitute expansion) and independent of the budgets then most of the results presented in this paper would all go through unchanged. It would be important to study in future work whether this is true and if not how the results change once this is taken into account.

A Proofs

A.1 Proof of Surjectivity Lemma 1

Suppose first, hypothetically, the global institution could make countries pay transfers and suppose it were to impose the reward payment scheme $\theta_{ixt}(\tilde{x}_{it} - x_{it})$. Then since $x \mapsto px - C_i(x) + \theta_{ix}(\tilde{x}_i - x)$ is concave, its global optimum is x_i iff $\theta_{ix} = p - C'_i(x_i)$. Now suppose the global institution offers instead the reward payment scheme $f_{ix}(x_i) = \max(0, \theta_{ixt}(\tilde{x}_{it} - x_{it}))$. x_{it} is still an optimal choice for i iff $px_i - C_i(x_i) + \theta_{ixt}(\tilde{x}_{it} - x_{it}) \geq \sup_x px - C_i(x)$, or equivalently:

$$\theta_{ixt}(\tilde{x}_{it} - x_{it}) \geq F_{ix} = \sup_x px - C_i(x) - (px_i - C_i(x_i))$$

Thus we have shown that the minimal transfer required to pay the country i to induce it to choose x_i under some affine linear reward payment scheme is indeed $F_{ix} = \sup_x px - C_i(x) - (px_i - C_i(x_i))$. Analogously, one can prove the claims for the other variables.

What is left to be proved is that there does not exist another set $(f_i)_{i \in I}$ of reward payment schemes implementing $((x_i, y_i, z_i)_{i \in I}, p)$ with a strictly smaller budget. To establish this, we note that the individual rationality condition implies:

$$B_i(y_i) - C_i(x_i) - G_i(z_i) + p(x_i - y_i + z_i) + f_i(x_i, y_i, z_i) \geq \sup_{(x,y,z)} B_i(y) - C_i(x) - G_i(z) + p(x - y + z) = \sup_y B_i(y) - py - \sup_x (C_i(x) - px) - \sup_z (G_i(z) - gz),$$

so we have:

$$f_i(x_i, y_i, z_i) \geq \sup_x px - C_i(x) - (px_i - C_i(x_i)) + \sup_y B_i(y) - py - (B_i(y_i) - py_i) + \sup_z pz - G_i(z) - (pz_i - G_i(z_i)) = F_{ix} + F_{iy} + F_{iz}.$$

A.2 Proof of Lemma 2

Proof. If the budget is sufficiently large, the global institution's problem's Lagrangian becomes equal to:

$$L = \sum_{j \in I} U_j - \eta(\sum_{j \in I} x_j) + \mu \sum_{i \in I} (x_i - y_i + z_i)$$

where μ is the Lagrange multiplier associated with the feasibility constraint.

The first order conditions are:

$$\frac{\partial L}{\partial x_i} = \frac{\partial U_i}{\partial x_i} - \eta + \mu = -C'_i(x_i) - \eta + \mu = 0$$

$$\frac{\partial L}{\partial y_i} = \frac{\partial U_i}{\partial y_i} - \eta + \mu = B'_i(y_i) - \mu = 0$$

$$\frac{\partial L}{\partial z_i} = \frac{\partial U_i}{\partial z_i} - \eta + \mu = -G'_i(z_i) + \mu$$

Thus the first best is characterized by the following conditions plus the feasibility condition $\sum_{i \in I} (x_i - y_i + z_i) = 0$:

$$B'_i(y_i) = B'_j(y_j) = G'_i(z_i) = G'_j(z_j) = C'_i(x_i) + \eta = C'_j(x_j) + \eta \forall i, j \in I$$

In particular, the world market price p of energy does not appear in this characterization. The greater the emphasis on rewarding countries for supply reduction, the larger the resulting price p will be. But as long as the required overall budget does not exceed the available budget, this does not matter for global welfare. $\square$

A.3 Proof of Price Preservation Lemma 3

Proof. Based on the Surjectivity Lemma (1), we view the global institution as choosing the price p and the allocation $(x_i, y_i, z_i)_{i \in I}$. Global welfare is determined by the allocation, $(x_i, y_i, z_i)_{i \in I}$, and the price p only is relevant because it affects the aggregate transfers required to get all countries to participate. The minimal required transfers $(F_{ix}, F_{iy}, F_{iz})_{i \in I}$ are given by the Surjectivity Lemma. In particular, the minimal aggregate required transfer is given by $\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}$.

We now show that $\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}$ is convex when viewed as a function of p . To do so, we use that $F_{ix} := \sup_x px - C_i(x) - (px_i - C_i(x_i))$, $F_{iy} := \sup_y B_i(y) - py - (B_i(y) - py)$, $F_{iz} := \sup_z pz - G_i(z) - (pz - G_i(z))$ and we compute:

$$\frac{d}{dp} (\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}) = \sum_{i \in I} x_i^*(p) - x_i - (y_i^*(p) - y_i) + z_i^*(p) - z_i$$

where $x_i^*(p)$ denotes country i 's supply function in the absence of any mechanism, i.e. $px - C_i(x) = \argmax_x px - C_i(x)$ and analogously for $y_i^*(p)$ and $z_i^*(p)$. By market clearing, we have:

$$\frac{d}{dp} (\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}) = \sum_{i \in I} x_i^*(p) - y_i^*(p) + z_i^*(p)$$

But this is just the excess supply function, which is strictly increasing in p as can be deduced directly from assumption 1. Thus the optimal p is characterized by the condition

$$\sum_{i \in I} x_i^*(p) - y_i^*(p) + z_i^*(p) = 0$$

The price obtaining at the market equilibrium in the absence of any mechanism satisfies this condition by market clearing. It is the unique price satisfying this condition. $\square$

A.4 Proof of Constrained Efficiency Lemma 4

Proof. The basic reason for this result is as follows: The global institution has two considerations to take into account: it cares intrinsically about the countries' aggregate utility and it wants to minimize the required transfers. But the outside options are determined by the price and so the required transfers decrease in the countries' aggregate utility. Thus the two considerations perfectly align. I will now flesh out this argument in full formal detail.

Consider a fixed price p . Consider the set S of all allocations satisfying the market clearing condition and having a given value X for $\sum x_i$.

$$S := \{(x_i, y_i, z_i) : \sum_{i \in I} (x_i - y_i + z_i) = 0, \sum_{i \in I} x_i = X\}$$

Let μ denote the Lagrange multiplier associated with the market clearing constraint. Let β denote the Lagrange multiplier associated with the global institution's budget constraint. The global institution's Lagrangian is:

$$L = \sum_{i \in I} U_i - \eta(\sum_{j \in I} x_j) + \mu \sum_{i \in I} (x_i - y_i + z_i) - \beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} - F)$$

When choosing amongst allocations in this set S , there are only two terms in the Lagrangian that are affected, namely $\sum_{i \in I} U_i$ and $-\beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz})$:

$$L = \sum_{i \in I} U_i - \eta(\sum_{j \in I} x_j) + \mu \sum_{i \in I} (x_i - y_i + z_i) - \beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} - F)$$

So we can write:

$$L = \sum_{i \in I} U_i - \beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}) + \phi$$

where ϕ does not depend on the allocation (as long as the allocation is chosen from the set S).

Using the expressions for the minimal transfers, we obtain:

$$L = \sum_{i \in I} U_i - \beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}) + \phi$$

We have:

$$\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} = \sup_x px - C_i(x) - (px_i - C_i(x_i)) + \sup_y B_i(y) - py - (B_i(y) - py) + \sup_z pz - G_i(z) - (py - G_i(y))$$

$$\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} = \sum_{i \in I} \sup_{x,y,z} p(x - y + z) - C_i(x) + B_i(y) - G_i(z) + C_i(x_i) - B_i(y_i) + G_i(z_i)$$

$$\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} = F + \sum_{i \in I} \sup_{x,y,z} p(x - y + z) - C_i(x) + B_i(y) - G_i(z) - U_i$$

Now we can write the Lagrangian as follows:

$$L = (1 + \beta) \sum_{i \in I} U_i + \phi^\#$$

where $\phi^\#$ does not depend on the allocation (as long as the allocation is chosen from the set S). Thus for any given p and a given value X for $\sum x_i$, the global institution's optimization problem for the allocation $(x_i, y_i, z_i)_{i \in I}$ boils down to simply maximizing $\sum_{i \in I} U_i$ under the constraint that $\sum x_i = X$. $\square$

A.5 Proof of the Interior Solution Corollary 1 and the Optimal Budget Split Corollary for Small Budgets 2

Proof. (of the Interior Solution Corollary and the Optimal Budget Split Corollary for Small Budgets) By the Price Preservation Lemma, the price at the optimal mechanism is the same as at the market equilibrium in the absence of any mechanism. We denote this price simply by p .

As before, let μ denote the Lagrange multiplier associated with the market clearing constraint. Let β denote the Lagrange multiplier associated with the global institution's budget constraint. The global institution's Lagrangian is:

$$L = \sum_{i \in I} U_i - \eta(\sum_{j \in I} x_j) + \mu \sum_{i \in I} (x_i - y_i + z_i) - \beta(\sum_{i \in I} F_{ix} + F_{iy} + F_{iz} - F)$$

By differentiating the Lagrangian with respect to the allocation, we obtain the following optimality conditions:

$$h_{xi}(x, y, z, \mu) := (1 + \beta)(p - C'_i(x_i)) + \mu - \eta' = 0$$

$$h_{yi}(x, y, z, \mu) := (1 + \beta)B'_i(y_i) - p - \mu = 0$$

$$h_{zi}(x, y, z, \mu) := (1 + \beta)(p - G'_i(z_i)) - p + \mu = 0$$

We also have the market clearing condition:

$$f_\mu(x, y, z, \mu) := \sum_i x_i - y_i + z_i = 0 \quad (1)$$

Define $\sigma := \frac{1}{1+\beta}$, so $\sigma = 0$ corresponds to the case where the budget $F = 0$ and $\sigma = 1$ corresponds to the case where the budget F is the minimal amount sufficient to implement the global optimum. With this, we can rewrite the optimality conditions as follows:

$$h_{xi}(x, y, z, \mu) := p - C'_i(x_i) + \sigma(\mu - \eta') = 0 \quad (2)$$

$$h_{yi}(x, y, z, \mu) := B'_i(y_i) - p - \sigma\mu = 0 \quad (3)$$

$$h_{zi}(x, y, z, \mu) := (1 + \beta)(p - G'_i(z_i)) - p + \sigma\mu = 0 \quad (4)$$

Now we can study what happens as we relax the budget constraint, which corresponds to increasing σ . Let us denote by $h := ((h_{xi})_{i \in I}, (h_{yi})_{i \in I}, (h_{zi})_{i \in I}, h_\mu)$, where h is a vector function whose components are defined in equations 2, 3, 4 and 1. The σ determines (x, y, z, μ) via the condition $h(x, y, z, \mu) = 0$.

For $\sigma \in (0, \infty)$ we know that this system has a unique solution. To see this, consider a fixed $\sigma > 0$. We can think of a choice of μ as determining the $(x_i, y_i, z_i)_{i \in I}$. Define $g(\mu) := h_\mu(x(\mu), y(\mu), z(\mu), \mu)$, where $x(\mu)$ denotes the vector of the x_i determined via equation 2 etc.. Each of the x_i and z_i are strictly increasing in μ , whilst all of the y_i are strictly decreasing in μ . Hence g is increasing. Moreover, $g(0) < 0$. To see why, we note that for $\mu = 0$ the y_i and the z_i are as in the absence of any mechanism whilst the x_i are strictly smaller. We also have that $g(\eta) > 0$. To see why, we note that for $\mu = \eta$ the x_i are as in the absence of any mechanism, whilst the y_i are strictly smaller and the z_i are strictly larger.

Given that thus $g(0) < 0$, $g(\eta) > 0$ and that $g(\mu)$ is increasing, we can apply the intermediate value theorem as long as g is continuous. But all the

$x_i(\mu), y_i(\mu), z_i(\mu)$ are continuous which implies that g is continuous. Hence, by the intermediate value theorem, there exists a unique μ such that equation 1 holds and this μ is in $(0, \eta)$. This shows that there is a unique solution to equations 2, 2, 4 and 1. We now denote the unique solution by $(x, y, z, \mu)(\sigma)$.

Now we will show that $(x, y, z, \mu)(\sigma)$ is continuously differentiable at all $\sigma \in (0, 1)$. For this it suffices by the implicit function theorem to show that the Jacobian of h is nonsingular on $(0, 1)$, since f is continuously differentiable. The Jacobian J is as follows:

$$J = \begin{pmatrix} -C''(x) & 0 & 0 & \sigma \\ 0 & B''(y) & 0 & -\sigma \\ 0 & 0 & -G''(z) & \sigma \\ 1 & -1 & 1 & 0 \end{pmatrix}$$

where $C''(x)$ is a diagonal matrix with entries $C''_i(x_i)$ etc. and in a slight abuse of notation the 6 zeros in the upper left denote $|I|$ by $|I|$ matrices with all entries 0 and the σ denote column vectors of length $|I|$. For $\sigma > 0$, J is non-singular, since by assumption $C''_i > 0, B''_i < 0, G''_i > 0$.

To see why, we note that if we want to write the bottom row vector as a linear combination of the other rows, then the weight given to each of the first $|I|$ rows (corresponding to the $(x_i)_{i \in I}$) must be strictly positive, whilst the weight given to the rows from row $|I| + 1$ to row $2|I|$ must be negative, whilst the weight given to the rows from row $2|I| + 1$ to row $3|I|$ must be negative. But all this together implies that the last component of this linear combination will be strictly positive, in contradiction to the fact that the last component of the last row is 0.

Hence we have established by the implicit function theorem that $(x, y, z, \mu)(\sigma)$ is continuously differentiable at all $\sigma \in (0, \infty)$. We also note that $\lim_{\sigma \rightarrow 0^+} J$ is singular. This explains why we will now need to do some more work to establish a fact that we will later need, namely that “generically” $\lim_{\sigma \rightarrow 0^+} \sigma \frac{d\mu}{d\sigma} = 0$.

Differentiation of the first order conditions (2,3,4) with respect to σ yields:

$$C''_i(x_i) \frac{dx_i}{d\sigma} = \mu - \eta + \sigma \frac{d\mu}{d\sigma} \quad (5)$$

$$B''_i(y_i) \frac{dy_i}{d\sigma} = \mu + \sigma \frac{d\mu}{d\sigma} \quad (6)$$

$$G''_i(z_i) \frac{dz_i}{d\sigma} = \mu + \sigma \frac{d\mu}{d\sigma} \quad (7)$$

Above we showed that $\mu(\sigma) \in (0, \eta) \forall \sigma > 0$. From this it follows that $\sigma \frac{d\mu}{d\sigma} \in (-\mu, \eta) \forall \sigma > 0$. This is because for σ such that $\sigma \frac{d\mu}{d\sigma} > \eta - \mu$ we would have $\frac{dx_i}{d\sigma} > 0 \forall i$ (by equation 5), $\frac{dy_i}{d\sigma} < 0 \forall i$ (by equation 6) and $\frac{dz_i}{d\sigma} > 0 \forall i$ (by equation 7), so that $\frac{d}{d\sigma}(x_i - y_i + z_i) > 0 \forall i$ which would contradict the market clearing condition.

Similarly, for $\sigma \frac{d\mu}{d\sigma} < -\mu$ we would have $\frac{dx_i}{d\sigma} < 0 \forall i$ (by equation 5), $\frac{dy_i}{d\sigma} > 0 \forall i$ (by equation 6) and $\frac{dz_i}{d\sigma} < 0 \forall i$ (by equation 7) so that $\frac{d}{d\sigma}(x_i - y_i + z_i) > 0 \forall i$, which would contradict the market clearing condition.

We are now ready to show that “generically” we must have $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma} = 0$. To establish this, let us first suppose that $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma}$ exists. Suppose we have $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma} = K \in (0, \infty]$. Then there exists some $\tilde{\sigma} > 0$ such that for all $\sigma \in (0, \tilde{\sigma}]$ we have $\frac{d\mu}{d\sigma} > \frac{K}{2\sigma}$. Integrating this yields for all $\sigma \in (0, \tilde{\sigma}]$: $\mu(\tilde{\sigma}) - \mu(\sigma) = \int_{\sigma}^{\tilde{\sigma}} \frac{d\mu}{ds} ds > \int_{\sigma}^{\tilde{\sigma}} \frac{K}{2s} ds = K(\log(\tilde{\sigma}) - \log(\sigma))$. Rearranging yields: $\mu(\sigma) < \mu(\tilde{\sigma}) - K(\log(\tilde{\sigma}) - \log(\sigma))$. But this would imply that $\lim_{\sigma \rightarrow 0} \mu(\sigma) = -\infty$, in contradiction to the fact that, as shown above, we always have $\mu(\sigma) \in (0, \eta) \forall \sigma$. Similarly, we can show that $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma} = K < 0$ leads to a contradiction.

Now in our quest to show that $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma} = 0$ we only have one more case to show to be impossible, namely the case where $\lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma}$ does not exist. But in this case $\frac{d\mu}{d\sigma}$ has to fluctuate infinitely often by unbounded amounts as σ approaches 0. Intuitively, this will “generically” never happen.

Now using the surjectivity lemma 1 yields:

$$\begin{aligned} \frac{dF_x}{d\sigma} &= \frac{d(\sum_i F_{ix})}{d\sigma} = \sum_i (C'_i(x_i) - p) \frac{\mu - \eta + \sigma \frac{d\mu}{d\sigma}}{C''_i(x_i)} \\ \frac{dF_y}{d\sigma} &= \frac{d(\sum_i F_{iy})}{d\sigma} = \sum_i -(B'_i(y_i) - p) \frac{\mu + \sigma \frac{d\mu}{d\sigma}}{B''_i(y_i)} \\ \frac{dF_z}{d\sigma} &= \frac{d(\sum_i F_{iz})}{d\sigma} = \sum_i (G'_i(x_i) - p) \frac{\mu + \sigma \frac{d\mu}{d\sigma}}{G''_i(x_i)} \end{aligned}$$

Using the optimality conditions yields:

$$\begin{aligned} \frac{dF_x}{d\sigma} &= \sum_i \sigma (\mu - \eta) \frac{\mu - \eta + \sigma \frac{d\mu}{d\sigma}}{C''_i(x_i)} \\ \frac{dF_y}{d\sigma} &= \sum_i \sigma \mu \frac{\mu + \sigma \frac{d\mu}{d\sigma}}{-B''_i(y_i)} \\ \frac{dF_{i,z}}{d\sigma} &= \sum_i \sigma \mu \frac{\mu + \sigma \frac{d\mu}{d\sigma}}{G''_i(x_i)} \end{aligned}$$

$$\frac{\frac{dF_x}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{(\mu - \eta + \sigma \frac{d\mu}{d\sigma})(\mu - \eta) \sum \frac{1}{C''_i(x_i)}}{(\mu - \eta + \sigma \frac{d\mu}{d\sigma})(\mu - \eta) \sum \frac{1}{C''_i(x_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{-B''_i(y_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{G''_i(x_i)}}$$

Now we differentiate the market clearing condition, getting

$$\sum \frac{dx_i}{d\sigma} + \frac{dz_i}{d\sigma} = \sum \frac{dy_i}{d\sigma}$$

Substituting into this gives:

$$(\mu - \eta + \sigma \frac{d\mu}{d\sigma}) \sum \frac{1}{C''_i(x_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \sum \frac{1}{G''_i(x_i)} = (\mu + \sigma \frac{d\mu}{d\sigma}) \sum \frac{1}{B''_i(y_i)}$$

$$\text{so } (\mu - \eta + \sigma \frac{d\mu}{d\sigma}) \sum \frac{1}{C''_i(x_i)} = -(\mu + \sigma \frac{d\mu}{d\sigma}) \left(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)} \right)$$

which we can plug in to get:

$$\begin{aligned} \frac{\frac{dF_x}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} &= \frac{-(\mu + \sigma \frac{d\mu}{d\sigma}) \left(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)} \right) (\mu - \eta)}{-(\mu + \sigma \frac{d\mu}{d\sigma}) \left(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)} \right) (\mu - \eta) + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{-B''_i(y_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{G''_i(x_i)}} \\ &= \frac{-(\mu + \sigma \frac{d\mu}{d\sigma}) \left(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)} \right) (\mu - \eta)}{-(\mu + \sigma \frac{d\mu}{d\sigma}) \left(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)} \right) (\mu - \eta) + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{-B''_i(y_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{G''_i(x_i)}} \\ &= \frac{-(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)}) (\mu - \eta)}{-(\sum \frac{1}{-B''_i(y_i)} + \sum \frac{1}{G''_i(x_i)}) (\mu - \eta) + \mu \sum \frac{1}{-B''_i(y_i)} + \mu \sum \frac{1}{G''_i(x_i)}} \\ &= \frac{-(\mu - \eta)}{-(\mu - \eta) + \mu} \\ &= \frac{\eta - \mu}{\eta} \end{aligned}$$

Increasing σ corresponds to increasing F . Therefore, we must have $\frac{dF_x}{d\sigma} +$

$\frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma} > 0$. But we already established that $\mu \in (0, \eta)$, so it follows that $\frac{dF_x}{d\sigma} > 0$.

We now proceed analogously for F_y :

$$\frac{\frac{dF_y}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{(\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{-B_i''(y_i)}}{(\mu + \sigma \frac{d\mu}{d\sigma}) (\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}) (\eta - \mu) + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{-B_i''(y_i)} + (\mu + \sigma \frac{d\mu}{d\sigma}) \mu \sum \frac{1}{G_i''(z_i)}}$$

$$\frac{\frac{dF_y}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{\mu \sum \frac{1}{-B_i''(y_i)}}{\eta (\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)})}$$

Hence $\frac{dF_y}{d\sigma} > 0$, since $\mu \in (0, \eta)$. $\square$

Now we proceed analogously to F_z :

$$\frac{\frac{dF_z}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{\mu \sum \frac{1}{G_i''(x_i)}}{\eta (\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)})}$$

Hence $\frac{dF_z}{d\sigma} > 0$, since $\mu \in (0, \eta)$.

Rearranging the condition derived from market clearing yields: $(\mu - \eta + \sigma \frac{d\mu}{d\sigma}) \sum \frac{1}{G_i''(x_i)} = -(\mu - \eta + \sigma \frac{d\mu}{d\sigma}) (\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}) - \eta (\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)})$

$$\mu - \eta = -\eta \frac{\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}}{\sum \frac{1}{G_i''(x_i)} + \sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}} - \sigma \frac{d\mu}{d\sigma}$$

$$\frac{\frac{dt_x}{d\sigma}}{\frac{dt_x}{d\sigma} + \frac{dt_y}{d\sigma} + \frac{dt_z}{d\sigma}} = \frac{\sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}}{\sum \frac{1}{G_i''(x_i)} + \sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}} + \frac{\sigma}{\eta} \frac{d\mu}{d\sigma}$$

using that $\mu_0 := \lim_{\sigma \rightarrow 0} \sigma \frac{d\mu}{d\sigma} = 0$ we get:

$$\lim_{\sigma \rightarrow 0} \frac{\frac{dt_x}{d\sigma}}{\frac{dt_x}{d\sigma} + \frac{dt_y}{d\sigma} + \frac{dt_z}{d\sigma}} = \frac{\sum \frac{1}{G_i''(x_i)} + \sum \frac{1}{-B_i''(y_i)}}{\sum \frac{1}{G_i''(x_i)} + \sum \frac{1}{-B_i''(y_i)} + \sum \frac{1}{G_i''(x_i)}}$$

Letting $y_i^*(p)$ denote as before the energy demand function for country i in the absence of any mechanism, we have:

$$B_i'(y_i^*(p)) = p, \text{ so } B_i''(p) \frac{dy_i^*}{dp}(p) = 1$$

Letting c denote the supply function for coal for country i in the absence of any mechanism, we have:

$$C_i'(x_i^*(p)) = p, \text{ so } C_i''(p) \frac{dx_i^*}{dp}(p) = 1$$

Letting z_i^* denote the supply function for coal for country i in the absence of any mechanism, we have:

$$G_i'(z_i^*(p)) = p(1 - q), \text{ so } G_i''(p) \frac{dz_i^*}{dp}(p) = 1$$

With this we obtain:

$\sum \frac{1}{-B_i''(y_i)} = \frac{dy_i^*}{dp} = \frac{y_i^*(p)}{p} \frac{p}{y_i^*(p)} \frac{dy_i^*}{dp} = \frac{y_i^*(p)}{p} \epsilon_y$, where ϵ_D denotes the price elasticity of demand for energy.

$\sum \frac{1}{C_i''(x_i)} = \frac{dx_i^*}{dp} = \frac{x_i^*(p)}{p} \frac{p}{x_i^*(p)} \frac{dx_i^*}{dp} = \frac{x_i^*(p)}{p} \epsilon_x$, where ϵ_x denotes the price elasticity of supply of coal.

$\sum \frac{1}{G_i''(z_i)} = \frac{dz_i^*}{dp} = \frac{z_i^*(p)}{p} \frac{p}{z_i^*(p)} \frac{dz_i^*}{dp} = \frac{z_i^*(p)}{p} \epsilon_{SG}$, where ϵ_{SG} denotes the price elasticity of supply of renewable energy.

Using these identities we get:

$$\lim_{\sigma \rightarrow 0} \frac{\frac{dF_x}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_D}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y} = \frac{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_y}{\epsilon_{SG} \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y}$$

where $X(0), Y(0), Z(0)$ denote the aggregate quantities for $F = 0$.

From this we deduce:

$$\lim_{\sigma \rightarrow 0} \frac{\frac{dF_y}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{\epsilon_y}{\epsilon_z + \epsilon_y} \frac{\epsilon_x \frac{X(0)}{Y(0)}}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y}$$

$$\lim_{\sigma \rightarrow 0} \frac{\frac{dF_z}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} = \frac{\epsilon_z}{\epsilon_z + \epsilon_y} \frac{\epsilon_x \frac{X(0)}{Y(0)}}{\epsilon_z \frac{Z(0)}{Y(0)} + \epsilon_x \frac{X(0)}{Y(0)} + \epsilon_y}$$

But we also have:

$$\frac{dF_x}{dF} = \frac{\frac{dF_x}{d\sigma}}{\frac{dF}{d\sigma}} = \frac{\frac{dF_y}{d\sigma}}{\frac{dF_x}{d\sigma} + \frac{dF_y}{d\sigma} + \frac{dF_z}{d\sigma}} \text{ so the claimed result follows. } \square$$

A.6 Proof of Lemma 5

Proof. Let $(x_i(F), y_i(F), z_i(F))$ be the optimal allocation under the budget F . Given any $F_1, F_2 > 0$, suppose we have a budget of $\phi F_1 + (1 - \phi)F_2$ with $\phi \in (0, 1)$. Set p equal to the status quo value in the absence of any mechanism and set $(x_i, y_i, z_i) = (\phi x_i(F_1) + (1 - \phi)x_i(F_2), \phi y_i(F_1) + (1 - \phi)y_i(F_2), \phi z_i(F_1) + (1 - \phi)z_i(F_2))$. The transfers required for this allocation to satisfy the participation constraints is lower than $\phi F_1 + (1 - \phi)F_2$ by the assumed convexity of C_i and G_i and the assumed concavity of B_i . Suppose we set the transfer so that the participation constraints are satisfied with equality. Then, denoting by $\tilde{U}_i$ the values of U_i under the status quo, we have:

$$\begin{aligned} W &= \sum \tilde{U}_i - \eta(\phi x_i(F_1) + (1 - \phi)x_i(F_2)) \\ &= \sum \phi(\tilde{U}_i - \eta x_i(F_1)) + (1 - \phi)(\tilde{U}_i - \eta x_i(F_2)) \\ &= \phi W(F_1) + (1 - \phi)W(F_2) \end{aligned}$$

But since we have not even used up all our budget, this shows that we can do strictly better than this. $\square$

A.7 Proof of Lemma 6

Proof. The first order conditions for the x_i require that $C'_i(x_i) = C'_j(x_j) \forall i, j$. Thus once we stipulate a value for $\sum_i x_i$, all the x_i are determined via $C'_i(x_i) = C'_j(x_j) \forall i, j$. Then all the y_i, z_i are determined via $\sum_i x_i = \sum_i y_i - \sum_i z_i$, $B'_i(y_i) = B'_j(y_j) = G'_i(z_i) = G'_j(z_j)$. Thus in particular, once $\sum_i x_i$ is fixed, the welfare W is determined and the transfers only depend on p . Conversely, W determines all the x_i, y_i, z_i and the F_x then only depends on p .

Specifically, F_x corresponds to p via $F_x = \sum_i F_{ix} = \sum_i p x_i^*(p) - C_i(x_i^*(p)) - (p x_i - C_i(x_i))$

where, as usual, we denote by $x_i^*(p)$ i 's coal supply in the absence of the mechanism.

$$\begin{aligned} \frac{dF_x}{dp} &= \sum_i x_i^* + (p - \alpha) \sum_i \frac{dx_i^*}{dp} - \sum_i \frac{dx_i^*}{dp} C'_i(x_i^*) = \sum_i x_i^*, \text{ so } \frac{dp}{dF_x} = \frac{1}{\sum_i x_i^*} \\ \text{Similarly, we get:} \\ \frac{d \sum_i F_{iy}}{dF_x} &= \frac{d \sum_i F_{iy}}{dp} \frac{dp}{dF_x} = \frac{-\sum_i y_i^*}{\sum_k x_k^*}, \frac{d \sum_i F_{iz}}{dF_x} = \frac{\sum_i z_i^*}{\sum_i x_i^*}, \text{ so } \frac{dF}{dF_x} = \frac{\sum_i x_i^* + \sum_i z_i^* - \sum_i y_i^*}{\sum_k x_k^*} \\ \frac{d^2 F}{dF_x^2} &= \frac{d}{dp} \frac{\sum_i x_i^* + z_i^* - y_i^*}{\sum_k x_k^*} \frac{dp}{dF_x} = \frac{-\sum_k \frac{dx_k^*}{dp} \sum_i (x_i^* + z_i^* - y_i^*) + \sum_k x_k^* (\sum_i \frac{dx_i^*}{dp} + \frac{dz_i^*}{dp} - \frac{dy_i^*}{dp})}{(\sum_k x_k^*)^2} \frac{dp}{dF_x} = \\ &= \frac{\sum_k \frac{dx_k^*}{dp} \sum_i (y_i^* - z_i^*) + \sum_k x_k^* (\frac{dz_i^*}{dp} - \frac{dy_i^*}{dp})}{(\sum_i x_i^*)^3} \end{aligned}$$

At the optimal mechanism, each country i will be paid to lower their energy use relative to what it would individually choose were it to ignore the mechanism.

Hence we must have $y_i \leq y_i^* \forall i$. Similarly, we must have $z_i \geq z_i^* \forall i$. Moreover, market clearing implies that $\sum_i y_i - z_i \geq 0$, so $\sum_i (y_i^* - z_i^*) \geq 0$.

Since $\frac{dz_i^*}{dp} \geq 0$, $\frac{dx_i^*}{dp} \geq 0 \forall i$ by the law of supply and $\frac{dy_i^*}{dp} \leq 0$ by the law of demand, it follows that $\frac{d^2 F}{dF_x^2} \geq 0$ $\square$

A.8 Proof of Lemma 8

Proof. Country i 's problem on the supply side is to choose a nondecreasing sequence of cumulative extraction $(x_{it})_{t=1,\dots,T}$ so as to maximize the discounted payoffs:

$$\begin{aligned} \frac{dU_i}{dx_{it}} &= \frac{1}{(1+r)^t} (p_t - \frac{1}{1+r} p_{t+1} - c'_{it}(x_{it}) + \frac{1}{1+r} c'_{it+1}(x_{it})) \\ \frac{d^2 U_i}{dx_{it}^2} &= \frac{1}{(1+r)^t} (-c''_{it}(x_{it}) + \frac{1}{1+r} c''_{it+1}(x_{it})) \end{aligned}$$

By assumption 6 we can from this conclude that $\frac{d^2 U_i}{dx_{it}^2} < 0 \forall t \in \{1, \dots, T-1\}$. Since $\frac{d^2 U_i}{dx_{it} dx_{jt}} \forall i \neq j$, we can conclude that the function U_i is concave in $(x_{it})_{t \in \{1, \dots, T\}}$ on R^T . It follows in particular that it is concave on the subspace of R^T defined by the condition that $(x_{it})_{t=1, \dots, T}$ be non-decreasing. Thus there exists a unique utility maximizing extraction path for country i . $\square$

A.9 Proof of Lemma 7

Proof. ($\implies$)

Suppose first that assumption 5 holds. Define $x_t := \sum_{i \in I} x_{it}$. Suppose we add emissions at time t . This will strictly increase $x_{t'}$ for all $t \geq t'$ whilst leaving x_t unchanged for $t' < t$. Hence η will strictly increase. This shows that property 1) holds. Now suppose we shift emissions from time t to time $t-1$. This will strictly increase x_{t-1} whilst leaving all $x_{t'}$ for $t' \neq t-1$ unchanged. Hence η will strictly increase.

($\impliedby$)

Suppose now that properties 1) and 2) hold. Suppose we increase x_t whilst holding all the other $x_{t'}$ for $t' \neq t$ constant. Suppose first that $t = T$. Then this consists of adding emissions in period T , so by property 1) this will strictly increase η . Now suppose that $t < T$. Then this consists of shifting emissions from period $t+1$ to period t . By property 2, this will strictly increase η . $\square$

A.10 Proof of Lemma 9

Proof. Formally, we can reinterpret the Walrasian World model of this paper as a classical Walrasian model with a numeraire good by viewing each country as being comprised of three parts: 1) a representative consumer with utility B_i that is quasilinear in the numeraire good¹⁷, 2) a firm able to use the numeraire good

¹⁷Of course, for the application of the model, the appropriate interpretation of the model is that B_i captures both the utility that consumers in i derive from energy consumption and the profits the benefits generated from energy for the production of other goods. The narrower interpretation adopted in this proof simply helps to use the results from standard Walrasian

to extract coal and turn it into energy, 3) as a firm able to use the numeraire good to produce renewable energy energy.

Now existence of a market equilibrium follows from proposition 17.BB.2 of Mas-Colell et. al 19.. (page 634), using assumption 4.

To establish uniqueness, we note that the first welfare theorem requires that any Walrasian equilibrium is Pareto optimal. However, given that all utilities are quasilinear in the same numeraire, this means that every Walrasian equilibrium must maximize the sum of all the utilities. But since all the utilities are concave functions of the allocations, this uniquely determines the allocation up to transfers in the numeraire good between consumers. $\square$

A.11 Proof of Surjectivity Lemma, Dynamic Version with full Commitment 10

Proof. For the demand side and the substitute side, the claim is established by the same arguments as in the proof of Lemma 1. On the supply side there is one complication to be dealt with: The constraint that the extraction sequence be non-decreasing. Country i 's problem on the supply side is to choose a non-decreasing sequence of cumulative extraction $(x_{it})_{t=1,\dots,T}$ so as to maximize the discounted payoffs:

$$\sum_{t=1}^T \frac{1}{(1+r)^t} (p_t(x_{it} - x_{it-1}) + f_{ixt}(x_{it}) - (c_{it}(x_{it}) - c_{it}(x_{it-1})))$$

Let us rewrite this by grouping together the terms involving a given x_{it} :

$$\sum_{t=1}^{T-1} \frac{1}{(1+r)^t} ((p_t - \frac{1}{1+r} p_{t+1}) x_{it} + f_{ixt}(x_{it}) - c_{it}(x_{it}) + \frac{1}{1+r} c_{it+1}(x_{it})) + \frac{1}{(1+r)^T} (p_T x_{iT} + f_{ixT}(x_{iT}) - c_{iT}(x_{iT})) \quad (8)$$

Suppose the global institution were to offer the following reward payment schemes: In period t pay the amount F_{ixt} to i if i chooses x_{it} and 0 otherwise, where F_{ixt} is defined as follows

$$F_{ixt} := (p_t - \frac{1}{1+r} p_{t+1})(x_{it}^*(p) - x_{it}) + (c_{it}(x_{it}) - c_{it}(x_{it}^*(p))) + \frac{1}{1+r} (c_{it+1}(x_{it}^*(p)) - c_{it+1}(x_{it})) \text{ for } t \in \{1, \dots, T-1\}$$

$$F_{ixT} := p_T(x_{iT}^*(p) - x_{iT}) + (c_{iT}(x_{iT}) - c_{iT}(x_{iT}^*(p)))$$

Recall that by definition 5, $(x_{it}^*(p))_{t \in \{1, \dots, T\}}$ denotes the unique utility maximizing extraction path for country i . By equation 8, $x_{it}^*(p)$ is the unique value that maximizes the term of the profit in which it appears. Thus we have $F_{ixt} \geq 0 \forall i, t$. Thereby we have established that the reward payment schemes we have defined are positive.

Consider the relaxed problem for country i where it ignores the constraint that $(x_{it})_{t=1,\dots,T}$ has to be non-decreasing. From equation 8 it follows that $(x_{it})_{t=1,\dots,T}$ is an optimum for i under the reward payment schemes defined above. But by assumption, this $(x_{it})_{t=1,\dots,T}$ is non-decreasing, so it is also an optimum of the actual problem that country i faces.

Now we need to show that every reward payment scheme inducing country i to choose $(x_{it})_{t=1,\dots,T}$ must end up paying to i at least the amount $F_{ix} :=$
theory.

$\sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{ixt}$ in discounted money on rewarding supply reduction. But for this we simply note that country i always has the option of ignoring all of the supply side reward payment schemes altogether. From this and equation 8 the claimed result follows.

Finally, let us show that positive affine linear reward payment schemes are sufficient to achieve any given supply side allocation with minimal transfers.

Let $V_{ixt}(x_{it-1}, (p_s)_{s \geq t}, (f_{ixs})_{s \geq t})$ denote the maximal discounted profit that country i can reap from coal extraction from period t onwards, given that its cumulative extraction before that is x_{it-1} and given the continuation world market price path $(p_s)_{s \geq t}$ and given the continuation path $(f_{ixs})_{s \geq t}$ of reward payment schemes.

The individual rationality condition for i to accept x_{iT} conditional on having accepted $(x_{i1}, \dots, x_{iT-1})$ is:

$$x_{it} = \operatorname{argmax}_x p_t(x - x_{it-1}) + f_{ixt}(x - x_{it-1}) - (c_{it}(x) - c_{it}(x_{it-1})) + \frac{1}{1+r} V_{ixt+1}(x, (p_s)_{s \geq t+1}, (f_{ixs})_{s \geq t+1})$$

The corresponding first order condition is:

$$p_t - \theta_{ixt} - c'_{it}(x_{it}) + \frac{1}{1+r} \frac{\partial V_{ixt+1}}{\partial x_{it}} = 0$$

Using the envelope theorem, we compute:

$$\frac{\partial V_{ixt+1}}{\partial x_{it}} = -p_{t+1} + c'_{it+1}(x_{it})$$

Hence the first order conditions are:

$$p_t - \theta_{ixt} - c'_{it}(x_{it}) + \frac{1}{1+r} (-p_{t+1} + c'_{it+1}(x_{it})) = 0 \text{ for } t \leq T-1$$

$$p_T - \theta_{ixT} - c'_{iT}(x_{iT}) = 0 \text{ for } t = T$$

Thus in each period t there is a unique θ_{it} such that x_{it} satisfies the first order condition in period t . What remains to be checked is that the second order conditions hold. For period T , this is true by since c_{iT} is convex by assumption. For periods $t < T$, the second order condition for a maximum is:

$$-c''_{it}(x_{it}) + \frac{1}{1+r} c''_{it+1}(x_{it}) \leq 0$$

But by assumption 6, this condition holds. In fact, assumption 6 implies that the function $x \mapsto p_t(x - x_{it-1}) + f_{ixt}(x - x_{it-1}) - (c_{it}(x) - c_{it}(x_{it-1})) + \frac{1}{1+r} V_{ixt+1}(x, (p_s)_{s \geq t+1}, (f_{ixs})_{s \geq t+1})$ is concave and thus the first order condition gives the unique maximum.

Adjusting the reference levels of the positive affine linear reward payment schemes simply amounts to varying the transfer that a country gets conditional on accepting the reward payment scheme (instead of ignoring it and maximizing its payoff without the rewards payments instead). Hence there are unique reference levels such that the the global institution ends up paying the minimal transfers. $\square$

A.12 Proof of Price Preservation Lemma, Dynamic Version with Full Commitment 11

Proof. By the Surjectivity Lemma 10, we can view the global institution as if it was choosing a world market price path $(p_t)_{t \in \{1, \dots, T\}}$ and an allocation $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$. Also by the Surjectivity Lemma the minimal discounted sum of transfers that the global institution has to pay for rewarding supply reduction is $F_{ix} := \sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{ixt}$ where F_{ixt} is defined as follows:

$$F_{ixt} := (p_t - \frac{1}{1+r} p_{t+1})(x_{it}^*(p) - x_{it}) + (c_{it}(x_{it}) - c_{it}(x_{it}^*(p))) + \frac{1}{1+r} (c_{it+1}(x_{it}^*(p)) - c_{it+1}(x_{it})) \text{ for } t \in \{1, \dots, T-1\}$$

$$F_{ixT} := p_T(x_{iT}^*(p) - x_{iT}) + (c_{iT}(x_{iT}) - c_{iT}(x_{iT}^*(p)))$$

Moreover, the minimal discounted sum of transfers that the global institution has to pay for inducing the demand reduction and the substitute expansion are $F_{iy} := \sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{iyt}$ and $F_{iz} := \sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{izt}$ with the following definitions:

$$F_{iyt} := \sup_y b_{it}(y) - p_t y - (b_{it}(y_{it}) - p_t y_{it})$$

$$F_{izt} := \sup_z p_t z - g_{it}(z) - (p_t z_{it} - g_{it}(z_{it}))$$

Let us use the envelope theorem to compute how the minimal value of discounted aggregate transfer is affected by the world market prices:

$$\frac{dF_{ix}}{dp_t} = \frac{1}{(1+r)^t} (x_{it}^*(p) - x_{it}) - \frac{1}{(1+r)^{t-1}} \frac{1}{1+r} (x_{it-1}^*(p) - x_{it-1})$$

$$\frac{dF_{ix}}{dp_t} = \frac{1}{(1+r)^t} (x_{it}^*(p) - x_{it-1}^*(p) - (x_{it} - x_{it-1}))$$

$$\frac{d}{dp_t} (\sum_{i \in I, t \in \{1, \dots, T\}} F_{ix} + F_{iy} + F_{iz}) = \sum_{i \in I} \frac{1}{(1+r)^t} (x_{it}^*(p) - x_{it-1}^*(p) - (x_{it} - x_{it-1}) - y_{it}^*(p) + y_{it} + z_{it}^*(p) - z_{it})$$

By market clearing in period t we deduce from this:

$$\frac{d}{dp_t} (\sum_{i \in I} F_{ix} + F_{iy} + F_{iz}) = \sum_{i \in I, t \in \{1, \dots, T\}} \frac{1}{(1+r)^t} (x_{it}^*(p) - x_{it-1}^*(p) - y_{it}^*(p) + z_{it}^*(p))$$

Optimality of the world market price path for the problem of minimising discounted aggregate transfers implies that this expression has to be 0 for each $t \in \{1, \dots, T\}$. But this set of conditions is equivalent to the statement that $((x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}, p)$ be a market equilibrium in the absence of any reward payment schemes. By Lemma 9 there is a unique such market equilibrium. $\square$

A.13 Proof of Constrained Efficiency Lemma, Dynamic Version 12

Proof. The global institution's objective is to maximize global welfare:

$$W = \sum_i U_i((x_{it}, y_{it}, z_{it})_{t=1, \dots, T}) - \eta(\sum_{i \in I} x_{i1}, \dots, \sum x_{iT})$$

under the constraint that the total amount of required transfers does not exceed the available intertemporal budget:

$$\sum_i \sum_{t=1, \dots, T} \frac{1}{(1+r)^t} F_{it} \leq F$$

Consider a fixed value for the total climate change damages, $\eta(\sum_{i \in I} x_{i1}, \dots, \sum x_{iT}) = \tilde{\eta}$. How should the global institution optimally choose the allocation $(x_{it}, y_{it}, z_{it})_{t=1, \dots, T}$ under the constraint that $\eta(\sum_{i \in I} x_{i1}, \dots, \sum x_{iT}) = \tilde{\eta}$? There are two considerations: The institution should try to reduce the transfers it needs to pay. For this it is best to maximize the countries' aggregate utility, $\sum_i U_i((x_{it}, y_{it}, z_{it})_{t=1, \dots, T})$, since the required transfers are determined so as to make up for the utility loss that countries occur relative to ignoring the reward payment schemes and optimizing given the market price vector p . The global institution also cares intrinsically about the countries utilities. Hence the two considerations perfectly align and the global institution should choose the allocation so as to maximize $\sum_i U_i((x_{it}, y_{it}, z_{it})_{t=1, \dots, T})$ under the constraint that $\eta(\sum_{i \in I} x_{i1}, \dots, \sum x_{iT}) = \tilde{\eta}$. $\square$

A.14 Proof of Monotone Mitigation Lemma 13

Proof. By the Constrained Efficiency Lemma 12 we know that the allocation at the optimal mechanism must maximize global welfare under the constraint that the eventual cumulative coal extraction be some fixed value X . For $X = \sum_i x_{iT}^*(p)$ the optimum is $x_{it} = x_{it}^*(p)$ by the first Welfare Theorem.

From now on we shall reason about how to optimally choose the $x_t := \sum_i x_{it}$, it being understood that the aggregate quantities are split up across countries so as to maximize global welfare. This is justified by the Constrained Efficiency Lemma 12. We use the definition of aggregate costs: $c_t(x_t) := \min_{(x_{it})_i} \sum c_{it}(x_{it})$.

Let us denote $(x_t(X), y_t(X), z_t(X))_{t \in \{1, \dots, T\}}$ the optimal allocation under the constraint that $x_T = X$. We shall now prove that $0 < \frac{dx_1}{dX} < \dots < \frac{dx_{T-1}}{dX} < 1$. For this, we note that by optimality we have:

$$\frac{\partial W}{\partial x_t} = 0 \forall t \leq T-1$$

Differentiating this with respect to X yields:

$$\frac{\partial^2 W}{\partial x_t \partial x_{t-1}} \frac{dx_{t-1}}{dX} + \frac{\partial^2 W}{\partial x_t^2} \frac{dx_t}{dX} + \frac{\partial^2 W}{\partial x_t \partial x_{t+1}} \frac{dx_{t+1}}{dX} = 0 \forall t \in \{2, \dots, T-2\}$$

$$\frac{\partial^2 W}{\partial x_1^2} \frac{dx_1}{dX} + \frac{\partial^2 W}{\partial x_1 \partial x_2} \frac{dx_2}{dX} = 0$$

$$\frac{\partial^2 W}{\partial x_{T-1}^2} \frac{dx_{T-1}}{dX} + \frac{\partial^2 W}{\partial x_{T-1} \partial X} = 0$$

Rearranging yields:

$$\frac{dx_{t+1}}{dX} = -\frac{\frac{\partial^2 W}{\partial x_t \partial x_{t-1}}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_{t-1}}{dX} - \frac{\frac{\partial^2 W}{\partial x_t^2}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_t}{dX} \forall t \in \{2, \dots, T-2\} \quad (9)$$

$$\frac{dx_2}{dX} = \frac{-\frac{\partial^2 W}{\partial x_1^2}}{\frac{\partial^2 W}{\partial x_1 \partial x_2}} \frac{dx_1}{dX} \quad (10)$$

$$\frac{dx_{T-1}}{dX} = \frac{\frac{\partial^2 W}{\partial x_{T-1} \partial x_T}}{-\frac{\partial^2 W}{\partial x_{T-1}^2}} \quad (11)$$

For better readability I will now assume that there is no substitute (i.e. no renewable energy). It is straightforward to generalise the proof that I will now give.

We have:

$$\frac{\partial W}{\partial x_t} = \frac{1}{(1+r)^t} (b'_t(x_t - x_{t-1}) - \frac{1}{1+r} b'_{t+1}(x_{t+1} - x_t) - c'_t(x_t) + \frac{1}{1+r} c'_{t+1}(x_{t+1}))$$

$$\frac{\partial^2 W}{\partial x_t^2} = \frac{1}{(1+r)^t} (b''_t(x_t - x_{t-1}) + \frac{1}{1+r} b''_{t+1}(x_{t+1} - x_t) - c''_t(x_t))$$

$$\frac{\partial^2 W}{\partial x_t \partial x_{t+1}} = \frac{1}{(1+r)^t} (-\frac{1}{1+r} b''_{t+1}(x_{t+1} - x_t))$$

In particular, we have:

$$0 < \frac{\partial^2 W}{\partial x_t \partial x_{t+1}} < \frac{\partial^2 W}{\partial x_t^2} \quad (12)$$

From 12 and 11 we deduce that $0 < \frac{dx_{T-1}}{dX} < 1$.

Let us now assume that $0 < \frac{dx_1}{dX}$, reserving the task of proving that this must hold for later. From 10 and 12 we deduce that $\frac{dx_1}{dX} < \frac{dx_2}{dX}$. Now we will prove that $\frac{dx_t}{dX} < \frac{dx_{t+1}}{dX} \forall t \in \{1, \dots, T-2\}$ by mathematical induction on t .

We have already established the base case with $t = 1$. Now we prove the induction step. For this, suppose that the claim holds for t , i.e suppose that $\frac{dx_t}{dX} > \frac{dx_{t-1}}{dX} > 0$. By 9, we have:

$$\frac{dx_{t+1}}{dX} = -\frac{\frac{\partial^2 W}{\partial x_t \partial x_{t-1}}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_{t-1}}{dX} - \frac{\frac{\partial^2 W}{\partial x_t^2}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_t}{dX}$$

Now using the induction hypothesis according to which $\frac{dx_t}{dX} > \frac{dx_{t-1}}{dX}$ and the fact $\frac{\partial^2 W}{\partial x_t \partial x_{t+1}} = \frac{1}{(1+r)^t} (-\frac{1}{1+r} b''_{t+1}(x_{t+1}-x_t)) > 0$ and $\frac{\partial^2 W}{\partial x_t \partial x_{t-1}} = \frac{1}{(1+r)^t} (-b''_t(x_t - x_{t-1})) > 0$, we deduce:

$$\frac{dx_{t+1}}{dX} > -\frac{\frac{\partial^2 W}{\partial x_t \partial x_{t-1}}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_{t-1}}{dX} - \frac{\frac{\partial^2 W}{\partial x_t^2}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_t}{dX} = \frac{-\frac{\partial^2 W}{\partial x_t^2} - \frac{\partial^2 W}{\partial x_t \partial x_{t-1}}}{\frac{\partial^2 W}{\partial x_t \partial x_{t+1}}} \frac{dx_t}{dX}$$

Substituting in yields:

$$\frac{dx_{t+1}}{dX} > \frac{-b''_t(x_t - x_{t-1}) - \frac{1}{1+r} b'_{t+1}(x_{t+1} - x_t) + c'_t(x_t) + b'_t(x_t - x_{t-1})}{-\frac{1}{1+r} b'_{t+1}(x_{t+1} - x_t)} \frac{dx_t}{dX} = \frac{-\frac{1}{1+r} b'_{t+1}(x_{t+1} - x_t) + c'_t(x_t)}{-\frac{1}{1+r} b'_{t+1}(x_{t+1} - x_t)} \frac{dx_t}{dX} >$$

Thus we have shown by induction that $\frac{dx_t}{dX} < \frac{dx_{t+1}}{dX} \forall t \in \{1, \dots, T-2\}$.

Now we will show that $0 \geq \frac{dx_1}{dX}$ would lead to a contradiction. In fact, the same inductive argument just provided would in this case imply that $\frac{dx_{T-1}}{dX} \leq \dots \leq \frac{dx_1}{dX} \leq 0$. But we already showed above that $0 < \frac{dx_{T-1}}{dX} < 1$.

Thus we have shown that $0 < \frac{dx_1}{dX} < \dots < \frac{dx_T}{dX} < 1$. In particular, we have $x_t(X) < \sum_i x_{it}^*(p) \forall t \forall W$ as long as $X < \sum_i x_{iT}^*(p)$. $\square$

A.15 Proof of the Time Inconsistency Corollary 6

Proof. Consider the situation at time T , assuming that to all countries' surprise at that time the global institution announces the new reoptimized mechanism. By the first part of the Monotone Mitigation Lemma 13, until time T less coal has been extracted than what would have been extracted in the absence of any mechanism. Let $p_T^\#$ denote the price that would occur if for the time period T no more mechanism was used and given that the cumulative coal extraction by the end of period $t-1$ was $(x_{iT-1})_{i \in I}$. Let $\hat{p}_T$ denote the price arising in period T if there was not any mechanism at any period. We have $p_T^\# < \hat{p}_T$, since otherwise there would be an excess coal supply under $p_T^\#$.

By the Dynamic Price Preservation Lemma 11, $\hat{p}_T$ is also the price arising under the optimal mechanism under full commitment. Thus if the global institution reneges on the announced mechanism at period T and reoptimizes, it will spend less on rewarding supply reduction. $\square$

A.16 Proof of the Price Rise Lemma 14

Proof. Let β_1, β_2 be the Lagrange multiplier associated with the budget constraint in the first and second periods, respectively. The Lagrangian writes as follows:

$$L = \sum_{i \in I} U_i - \eta (\sum_{j \in I} x_{j1}, \sum_{j \in I} x_{j2}) + \mu_1 \sum_{i \in I} (x_{i1} - y_{i1} + z_{i1}) + \mu_2 \sum_{i \in I} (x_{i2} - y_{i2} + z_{i2}) - \beta_1 (F_{1x} + F_{1y} + F_{1z} - F_1) - \frac{\beta_2}{1+r} (F_{2x} + F_{2y} + F_{2z} - F_2)$$

The no-borrowing constraint being binding means that $\beta_1 > \beta_2$: An additional dollar available at period 1 is more valuable than saving it for period 2, earning interest and spending it then.

$$\begin{aligned} \frac{\partial F_{x1}}{\partial p_1} &= \sum_i (x_{i1}^*(p) - x_{i1}) \\ \frac{\partial F_{x2}}{\partial p_2} &= \sum_i (x_{i2}^*(p) - x_{i2}) \\ \frac{\partial F_{x1}}{\partial p_2} &= -\frac{1}{1+r} \sum_i (x_{i1}^*(p) - x_{i1}) \\ \frac{dL}{dp_2} &= \beta_1 \frac{1}{1+r} \sum_i (x_{i1}^*(p) - x_{i1}) - \frac{\beta_2}{1+r} (\sum_i (x_{i2}^*(p) - x_{i2}) + \sum_i (z_{i2}^*(p) - z_{i2}) - \sum_i (y_{i2}^*(p) - y_{i2})) \end{aligned}$$

This we can rewrite as follows:

$$\frac{dL}{dp_2} = (\frac{\beta_1}{1+r} - \frac{\beta_2}{1+r}) \sum_i (x_{i1}^*(p) - x_{i1}) - \frac{\beta_2}{1+r} (\sum_i (x_{i2}^*(p) - x_{i1}^*(p)) - (x_{i2} - x_{i1})) + \sum_i (z_{i2}^*(p) - z_{i2}) - \sum_i (y_{i2}^*(p) - y_{i2}))$$

Now using market clearing we obtain:

$$\frac{dL}{dp_2} = (\frac{\beta_1}{1+r} - \frac{\beta_2}{1+r}) \sum_i (x_{i1}^*(p) - x_{i1}) - \frac{\beta_2}{1+r} (\sum_i (x_{i2}^*(p) - x_{i1}^*(p)) + \sum_i z_{i2}^*(p) - \sum_i y_{i2}^*(p))$$

At the optimal mechanism (under the no-borrowing constraint but assuming full commitment) we clearly have that $\sum_i (x_{i1}^*(p) - x_{i1}) > 0$, since otherwise spending in the first would be counterproductive. Thus, given that $\frac{dL}{dp_2} = 0$ implies that there would be excess supply of coal in the last period under the world market price path p and without reward payment schemes:

$$\sum_i (x_{i2}^*(p) - x_{i1}^*(p)) + \sum_i z_{i2}^*(p) - \sum_i y_{i2}^*(p) = (\frac{\beta_1}{\beta_2} - 1) \sum_i (x_{i1}^*(p) - x_{i1}) > 0 \quad (13)$$

We also have

$$\frac{dL}{dp_1} = -\beta_1 (\sum_i (x_{i1}^*(p) - x_{i1}) + \sum_i (z_{i1}^*(p) - z_{i1}) - \sum_i (y_{i1}^*(p) - y_{i1}))$$

Using market clearing yields:

$$\frac{dL}{dp_1} = -\beta_1 (\sum_i x_{i1}^*(p) + z_{i1}^*(p) - y_{i1}^*(p))$$

Thus by optimality of p we have:

$$\sum_i x_{i1}^*(p) + z_{i1}^*(p) - y_{i1}^*(p) = 0 \quad (14)$$

Thus the world market price path corresponding to the optimal mechanism would (if it were not for the reward payment schemes) lead to excess supply in the second period and market clearing in the first period.

Recall that the first order condition characterizing $x_{it}^*(p)$ is:

$$p_t - c'_{it}(x_{it}) + \frac{1}{1+r}(-p_{t+1} + c'_{it+1}(x_{it})) = 0$$

Hence

$$\frac{dx_{it}^*}{dp_t} = \frac{1}{c''_{it}(x_{it}^*(p))} > 0 \quad (15)$$

$$\frac{dx_{it}^*}{dp_{t+1}} = \frac{-1}{c''_{it}(x_{it}^*(p))(1+r)} < 0 \quad (16)$$

Suppose we had $p_2 < \hat{p}_2$. Then by equation 14 and 16 we would need to have $p_1 < \hat{p}_1$. But then we would have:

$$x_{1i}^*(p) > x_{1i}^*(\hat{p}) \text{ (by 14)}$$

$$x_{2i}^*(p) < x_{2i}^*(\hat{p}) \text{ (by 15)}$$

But this would imply:

$$\sum_i (x_{i2}^*(p) - x_{i1}^*(p)) < \sum_i (x_{i2}^*(\hat{p}) - x_{i1}^*(\hat{p}))$$

But since by assumption $p_2 < \hat{p}_2$, we would also have $z_{i2}^*(p) < z_{i2}^*(\hat{p}) \forall i, y_{i2}^*(p) > y_{i2}^*(\hat{p}) \forall i$. From this we would conclude that $\sum_i (x_{i2}^*(p) - x_{i1}^*(p)) + \sum_i z_{i2}^*(p) - \sum_i y_{i2}^*(p) < 0$, in contradiction to equation 13. Thus we have shown that $p_2 \leq \hat{p}_2$ would lead to a contradiction. Hence we must have $p_2 > \hat{p}_2$.

Now equations 14 and 16 imply that $p_1 > \hat{p}_1$ □

A.17 Proof of the Surjectivity Lemma, Dynamic Version without commitment 16

Proof. In period $t + 1$ the vector $(x_{it})_{i \in I}$ of the countries' cumulative coal extraction until then will be all that will matter going forward, given that by assumption the global institution cannot commit and cannot borrow or save. In particular, $(x_{it})_{i \in I}$ will determine the world market price path $(p_s)_{s \geq t+1}$. Since the global institution cannot commit, it will clearly never pay out rents. Thus in period t each country i can reason as if it does not get offered any reward payment schemes by the global institution in the future.

Given these observations, the claims now all follow from the reasoning provided in the proof of Lemma 10 □

A.18 Proof the Monotone Mitigation Lemma 17 and the Price Drop Lemma 18

Proof. Since the global institution cannot commit, it will clearly never pay out rents. Thus in period t each country i can reason as if it does not get offered any reward payment schemes by the global institution in the future. All that matters for country i is thus the future price path. In particular, in period 1, all that matters for country i is p_1, p_2 and f_{1i} . By the static version of the Price Preservation Lemma applied to period 2, F_2 does not affect p_2 . By the Surjectivity Lemma 16, we can view the global institution as facing in period 1 the problem of choosing p_1 and $(x_{1i})_{i \in I}$. Now since we have shown that F_2 does not affect p_2 , we can deduce that the global institution's problem in period 1

is unaffected by F_2 . In particular, the optimal solution to this problem will be unaffected by F_2 . It follows that $X_1(F_1, F_2) = X_1(F_1, 0) \forall F_1, F_2$.

Given that the global institution's problem in period 1 is unaffected by F_2 , the claimed statement " $X_1(F_1, F_2)$ is strictly decreasing in F_1 " will follow if we prove the statement " $X_1(F_1, 0)$ is strictly decreasing in F_1 ". But this is clearly true: the aggregate coal extraction by the end of period 1 is all that will matter going forward and by the Surjectivity Lemma 16 we can view the global institution as directly deciding on the $(x_{i1})_{i \in I}$ in period 1. With more money the global institution can (and thus will) achieve lower cumulative coal extraction.

Now let us study the world market prices. The Static Price Preservation Lemma applied to the period 2 immediately implies that $p_2(F_1, F_2) = p_2(F_1, 0) \forall F_1, F_2$.

Moreover, as we noted above, F_2 does not affect the global institution's problem in period 1, so in particular it will not affect the resulting first period price, so we get $p_1(F_1, F_2) = p_1(F_1, 0) \forall F_1, F_2$.

We have already shown that increasing F_1 will decrease X_1 . If $F_2 = 0$ then this will clearly lead to a decrease in p_2 . But since F_2 does not affect p_2 by the static Price Preservation Lemma, this implies that $p_2(F_1, F_2)$ is strictly decreasing in F_1 .

Recall from the Surjectivity Lemma 16 that the required transfers for rewarding supply reduction in the first period are:

$$F_{ix1} = (p_1 - \frac{1}{1+r}p_2)(x_{i1}^*(p) - x_{i1}) + (c_{i1}(x_{i1}) - c_{i1}(x_{i1}^*(p))) + \frac{1}{1+r}(c_{i2}(x_{i1}^*(p)) - c_{i2}(x_{i1}))$$

or equivalently:

$$F_{ix1} = \sup_x (p_1 - \frac{1}{1+r}p_2)x - c_{i1}(x) + \frac{1}{1+r}c_{i2}(x) - ((p_1 - \frac{1}{1+r}p_2)x_{i1} - c_{i1}(x_{i1}) + \frac{1}{1+r}c_{i2}(x_{i1}))$$

By the envelope theorem, we obtain:

$$\frac{dF_{ix1}}{dp_1} = x_{i1}^*(p) - x_{i1}$$

Similarly, we obtain:

$$\frac{dF_{iy1}}{dp_1} = y_{i1}^*(p) - y_{i1}$$

$$\frac{dF_{iz1}}{dp_1} = z_{i1}^*(p) - z_{i1}$$

Optimality requires that F_1 be minimal, so we must have:

$$\frac{dF_1}{dp_1} = \sum_{i \in I} x_{i1}^*(p) - x_{i0} - (y_{i1}^*(p) - y_{i1}) + z_{i1}^*(p) - z_{i1} = 0$$

But by market clearing we have $\sum_{i \in I} x_{i1} - x_{i0} - y_{i1} + z_{i1} = 0$, so we obtain:

$$\sum_{i \in I} x_{i1}^*(p) - x_{i0} - y_{i1}^*(p) + z_{i1}^*(p) = 0$$

Now consider what happens when we increase F_1 . We have already shown that this will decrease p_2 . But the only way that $\sum_{i \in I} x_{i1}^*(p) - x_{i0} - y_{i1}^*(p) + z_{i1}^*(p) = 0$ can still hold is if concomittantly p_1 has decreased. To see why this is so, consider country i 's first order condition (in the absence of reward payment schemes) in period 1:

$$p_1 - c'_{i1}(x_{i1}^*(p)) + \frac{1}{1+r}(-p_2 + c'_{i2}(x_{i1}^*(p))) = 0$$

If the decrease in p_2 were accompanied with an increase in p_1 , then $x_{i1}^*(p)$ would have increased. Moreover, $y_{i1}^*(p)$ would have decreased and $z_{i1}^*(p)$ increased. Together, this would contradict that all along we must have $\sum_{i \in I} x_{i1}^*(p) -$

$$x_{i0} - y_{i1}^*(p) + z_{i1}^*(p) = 0. \quad \square$$

A.19 Proof of the Endogenous Week Green Paradox Corollary for T=3 9

Proof. By the Price Drop Lemma 18, we know that increasing F_2 will strictly decrease p_2 conditional on any given $(x_{i1})_{i \in I}$. But recall from the Surjectivity Lemma 16 that the required transfer in period 1 to get i to choose x_{i1} is given by:

$$F_{ix1} = \sup_x (p_1 - \frac{1}{1+r} p_2) x - c_{i1}(x) + \frac{1}{1+r} c_{i2}(x) - ((p_1 - \frac{1}{1+r} p_2) x_{i1} - c_{i1}(x_{i1}) + \frac{1}{1+r} c_{i2}(x_{i1}))$$

Hence $\frac{dF_{ix1}}{dp_2} = -\frac{1}{1+r} (x_{1i}^*(p) - x_{i1}) < 0$, so the decrease in p_2 always increases the transfers required in period 1 to achieve a given $(x_{i1})_{i \in I}$. Hence with the fixed budget in period 1 less supply reduction can be achieved. $\square$

A.20 Proof of the Monotone Eventual Emissions Corollary 1

Proof. Suppose that (F_1, F_2, F_3) is such that the global institution's budget constraint is binding in each period. We have $\frac{\partial x_3}{\partial F_3} < 0$ by Proposition 2 applied to the last period. We also clearly have $\frac{\partial x_3}{\partial F_1} < 0$. Thus the only claim to be proven is that $\frac{\partial x_3}{\partial x_2} < 0$. This claim is non-obvious, given corollary 9. We will prove it by computing $\frac{\partial x_3}{\partial x_2}$ explicitly, a task that we now turn to.

By the Constrained Efficiency Lemma we can focus on the aggregate quantities in each period, it being understood that the distribution of e.g. extraction across countries is done so as to maximize aggregate welfare. We are interested in the mapping $(F_1, F_2, F_3) \mapsto (x_1, x_2, x_3)$ giving us aggregate cumulative extraction as a function of the funding stream, given that the global institution cannot commit or borrow or save.

Let us compute the inverse of this mapping. In other words, let us ask: If (x_1, x_2, x_3) is the outcome, what must have been the funding stream resulting in this? This will be useful because we can later use the Inverse Function Theorem to compute the derivative of the original mapping $(F_1, F_2, F_3) \mapsto (x_1, x_2, x_3)$ that we are interested in.

We note that at each period t the price p_t must be such that F_t is minimal under the constraint that x_t is to be implemented.

$$F_{xt} := (p_t - \frac{1}{1+r} p_{t+1})(x_t^*(p) - x_t) + (c_t(x_t) - c_t(x_t^*(p))) + \frac{1}{1+r} (c_{t+1}(x_t^*(p)) - c_{t+1}(x_t)) \text{ for } t \in \{1, \dots, T-1\}$$

$$F_{xT} := p_T(x_T^*(p) - x_T) + (c_T(x_T) - c_T(x_T^*(p)))$$

$$\frac{dF_{xt}}{dp_t} = x_t^*(p) - x_t \forall t$$

Similarly, we obtain:

$$\frac{dF_{yt}}{dp_t} = -y_t^*(p) + y_t$$

$$\frac{dF_{zt}}{dp_t} = z_t^*(p) - z_t$$

$$\text{Hence } \frac{dF_t}{dp_t} = x_t^*(p) - x_t - y_t^*(p) + y_t + z_t^*(p) - z_t$$

By market clearing we have $x_t - x_{t-1} - y_t + z_t = 0$. Thus we get:

$$\frac{dF_t}{dp_t} = x_t^*(p) - x_{t-1} - y_t^*(p) + z_t^*(p) \forall t$$

Minimality of F_t requires:

$$x_t^*(p) - x_{t-1} - y_t^*(p) + z_t^*(p) = 0 \forall t \in \{1, \dots, T\}$$

We also have:

$$p_T - c'_T(x_T^*(p)) = 0$$

$$p_t - c'_t(x_t^*(p)) + \frac{1}{1+r}(-p_{t+1} + c'_{t+1}(x_{t+1}^*(p))) = 0 \forall t \in \{1, \dots, T\}$$

Let us drop in the notation the dependence of x^* on p and let us view the equations obtained thus far as a system of $3T$ equations in the $3T$ unknowns,

$$((x_t^*)_{t \in \{1, \dots, T\}}, (y_t^*)_{t \in \{1, \dots, T\}}, (p_t)_{t \in \{1, \dots, T\}}):$$

$$x_t^* - x_{t-1} - y_t^* + z_t^* = 0 \forall t \in \{1, \dots, T\}$$

$$p_t - c'_t(x_t^*) + \frac{1}{1+r}(-p_{t+1} + c'_{t+1}(x_{t+1}^*)) = 0 \forall t \in \{1, \dots, T-1\}$$

$$p_T - c'_T(x_T^*) = 0$$

$$p_t - b'_t(y_t^*) = 0 \forall t \in \{1, \dots, T\}$$

This set of equations allows us to compute $(x_1^*, x_2^*, x_3^*), (y_1^*, y_2^*, y_3^*), (p_1, p_2, p_3)$ from (x_1, x_2, x_3) . By market clearing, we also have $(y_1, y_2, y_3) = (x_1, x_2, x_3)$.

All this together gives us the following factorization:

$$(x_1, x_2, x_3) \mapsto ((x_1, x_2, x_3), (y_1, y_2, y_3), (x_1^*, x_2^*, x_3^*), (y_1^*, y_2^*, y_3^*), (p_1, p_2, p_3)) \mapsto (F_1, F_2, F_3)$$

We can use this via the chain rule to compute the Jacobian of the mapping $(x_1, x_2, x_3) \mapsto (F_1, F_2, F_3)$. By the Inverse Function Theorem, the inverse of the Jacobian of this mapping is the Jacobian of the inverse mapping, which is what we are ultimately interested in. I computed all this with Mathematica. The documented Mathematica notebook is available for download [here](#). The algebraic expressions computed by Mathematica are very long. I now introduce some notation to make the reasoning readable.

Let us define:

$$q_{xt} := p_t - c'_t(x_t) + \frac{1}{1+r}(-p_{t+1} + c'_{t+1}(x_{t+1})) \text{ for } t \in \{1, \dots, T-1\}$$

$$q_{xT} := p_T - c'_T(x_T)$$

$$q_{yt} := b'_t(y_t) - p_t \text{ for } t \in \{1, \dots, T\}$$

From the Interior Solution Corollary 1 we deduce that $q_{xt} > 0, q_{yt} > 0 \forall t \in \{1, \dots, T\}$.

$$\text{Let us denote by } e_{xt} := \frac{dx_t^*}{dp_t} \frac{p_t}{x_t^*} \text{ and } e_{yt} := -\frac{dy_t^*}{dp_t} \frac{p_t}{y_t^*}.$$

In the Mathematica notebook I find that $\frac{\partial x_3}{\partial F_2} = \frac{a_1 a_2 a_3}{a_4 a_5}$ with the following definitions:

$$a_1 = q_{y3}(r+1)(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1))^2$$

$$a_2 = (r+1)(q_{x1} + q_{y1})(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1)) + p_2(x_1^* - x_1)$$

$$a_3 = -e_{x3}(x_3^* - x_2^*) - e_{y3}(x_3^* - x_2)$$

$$a_4 = (q_{x3} + q_{y3})(e_{x2}(x_1^* - x_2^*) + e_{y2}(x_2^* - x_1))$$

$$a_5 = a_{51} + a_{52}$$

$$a_{51} = p_3(a_{511} + a_{512})$$

$$a_{511} = e_{x2}q_{y2}(x_1^* - x_1)(x_1^* - x_2^*)(e_{x2}(x_1^* - x_2^*) - e_{y2}(x_2^* - x_1))$$

$$a_{512} = (x_2 - x_2^*)(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1))((r+1)(q_{x1} + q_{y1})(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1)) + p_2(x_1^* - x_1))$$

$$a_{52} = (r+1)(-(q_{x2} + q_{y2}))(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1))(e_{x3}(x_3^* - x_2^*) + e_{y3}(x_2 - x_3^*))((r+1)(q_{x1} + q_{y1})(e_{x2}(x_2^* - x_1^*) + e_{y2}(x_2^* - x_1)) + p_2(x_1^* - x_1))$$

We can now deduce:

$$a_1 > 0, a_2 > 0, a_3 < 0, a_4 < 0$$

Given this, establishing the claimed result now comes down to showing that $a_5 < 0$. To do so, we note that $a_{511} < 0$ and $a_{512} < 0$, so $a_{51} < 0$. We also have $a_{52} < 0$, so it follows that $a_5 < 0$

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□

A.21 Proof of Lemma 19

Proof. For infinitesimal budgets, any changes in aggregate private utility that the global institution induces are of second order, given that aggregate utility is maximal by the First welfare theorem. Thus changes in aggregate welfare are equal in changes to the aggregate external effects, i.e. η . Additional money in the first period cannot create any adverse effects on global welfare, so $\frac{d\eta}{dF_1}|_{(F_1, F_2, F_3)=0} \geq 0$. Additional money in the last period will not affect the price of energy in the last period, so this will not affect prior periods. Hence $\frac{d\eta}{dF_3}|_{(F_1, F_2, F_3)=0} \geq 0$. □

A.22 Proof of Lemma 10

Proof. As in proposition 2, let us define $a := e_{y1} \frac{1}{(\frac{x_3-x_2}{x_1})(1+\frac{e_{y1}}{e_{x1}})(1+\frac{e_{y2}}{e_{x2}}(\frac{x_2-x_1}{x_2}))\frac{p_1(1+r)}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{x_1}{x_2}\frac{p_2}{p_3}}$.

By proposition 2, the Strong Green Paradox arises for small budgets iff both of the following two conditions hold:

$$e_{y3} < a$$

$$e_{x3} < e_{y1} \frac{\frac{a-e_{y3}}{a}r}{\frac{x_3}{x_2}((1+\frac{e_{y1}}{e_{x1}})(\frac{x_2-x_1}{x_1} + \frac{x_2}{x_1}\frac{e_{y2}}{e_{x2}})\frac{p_1(1+r)^2}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{p_2(1+r)}{p_3})}$$

Suppose we want the Strong Green Paradox to arise. we want $a = e_{y1} \frac{1}{(\frac{x_3-x_2}{x_1})(1+\frac{e_{y1}}{e_{x1}})\frac{p_1(1+r)}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{x_1}{x_2}\frac{p_2}{p_3}}$

to be large and $e_{y1} \frac{\frac{a-e_{y3}}{a}r}{\frac{x_3}{x_2}((1+\frac{e_{y1}}{e_{x1}})(\frac{x_2-x_1}{x_1} + \frac{x_2}{x_1}\frac{e_{y2}}{e_{x2}})\frac{p_1(1+r)^2}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{p_2(1+r)}{p_3})}$ to be large. For both it is best to set $e_{y2} = 0$. With this we get the two conditions from proposition 2:

$$e_{y3} < a := e_{y1} \frac{1}{(\frac{x_3-x_2}{x_1})(1+\frac{e_{y1}}{e_{x1}})\frac{p_1(1+r)}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{x_1}{x_2}\frac{p_2}{p_3}}$$

$$e_{x3} < e_{y1} \frac{\frac{a-e_{y3}}{a}r}{\frac{x_3}{x_2}((1+\frac{e_{y1}}{e_{x1}})(\frac{x_2-x_1}{x_1})\frac{p_1(1+r)^2}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{p_2(1+r)}{p_3})}$$

Now if we want to get a Strong Green Paradox, it is always best to set $e_{y3} = 0$, which gives us:

$$e_{x3} \geq e_{y1} \frac{r}{\frac{x_3}{x_2}((1+\frac{e_{y1}}{e_{x1}})(\frac{x_2-x_1}{x_1})\frac{p_1(1+r)^2}{p_3} + \frac{e_{y1}}{e_{x2}}\frac{p_2(1+r)}{p_3})}$$

Now if we want to get a Strong Green Paradox, it is always best to set $e_{y1} = \infty$, which gives us:

¹⁸From this computation we can also heuristically deduce the following result: $\lim_{(F_1, F_2, F_3) \rightarrow 0} \frac{\partial X_3}{\partial F_2}|_{(F_1, F_2, F_3)} = -\infty$ and $\lim_{(F_1, F_2, F_3) \rightarrow 0} \frac{\partial W}{\partial F_2}|_{(F_1, F_2, F_3)} = \infty$.

$$e_{x3} \geq \frac{r}{\frac{x_3}{x_2} \left(\left(\frac{1}{e_{x1}} \right) \left(\frac{x_2 - x_1}{x_1} \right) \frac{p_1(1+r)^2}{p_3} + \frac{1}{e_{x2}} \frac{p_2(1+r)}{p_3} \right)}$$

This we can rewrite as follows:

$$\left(\left(\frac{1}{e_{x1}} \right) \left(\frac{x_2 - x_1}{x_1} \right) \frac{p_1(1+r)^2}{p_3} + \frac{1}{e_{x2}} \frac{p_2(1+r)}{p_3} \right) \geq \frac{r}{e_{x3}} \frac{x_2}{x_3}$$

$$\frac{p_1}{e_{x1} x_1} \left(\frac{x_2 - x_1}{x_2} \right) \frac{(1+r)^2}{r} + \frac{p_2}{e_{x2} x_2} \frac{1+r}{r} \geq \frac{p_3}{e_{x3} x_3}$$

Now differentiating the first order condition for $x_t^*(p)$, $p_t - c'_t(x_t^*(p)) + \frac{1}{1+r}(-p_{t+1} + c'_{t+1}(x_t^*(p))) = 0$, yields:

$$1 - c''_t(x_t^*(p)) \frac{dx_t^*}{dp_t} + \frac{1}{1+r} c''_{t+1}(x_t^*(p)) \frac{dx_t^*}{dp_t} = 0$$

$$\frac{dx_t^*}{dp_t} = \frac{1}{c''_t(x_t^*(p)) - \frac{1}{1+r} c''_{t+1}(x_t^*(p))}$$

Substituting this in gives u:

$$(c''_1(x_1) - \frac{1}{1+r} c''_2(x_1)) \left(\frac{x_2 - x_1}{x_2} \right) \frac{(1+r)^2}{r} + (c''_2(x_2) - \frac{1}{1+r} c''_3(x_2)) \frac{1+r}{r} \geq c''_3(x_3)$$

Now suppose that extraction costs do not change with time. Then we get:

$$c''(x_1) \left(\frac{x_2 - x_1}{x_2} \right) (1+r) + c''(x_2) \geq c''(x_3) \quad \square$$

A.23 The backward induction solution for the case with a single combined optimized fund for rewarding supply reduction and demand reduction 21

As reported in section 7.3.2, it turns out that all along the backward induction solution the aggregate contributions to the combined compensation fund will never exceed the aggregate contributions that arise at the static Nash equilibrium in the hypothetical one shot game where both players decide on their contributions simultaneously. The numerical results are rather messy. Hence instead of reporting them, I provide two Lemmas and simpler example with an explicit solution to explain the results.

Lemma 21. *Consider the game where there is only one combined fund for rewarding both supply reduction and demand reduction. Moreover, suppose that this fund uses any given budget so as to maximize global welfare¹⁹ (not taking into account that funding is actually endogenous). Then at no point of the backward induction solution in the game does any player's strategy increase in the other player's current contribution. Moreover, this claim also holds at an Markov Perfect Equilibrium.*

Proof. Player 1's payoff can be written as $\theta_1(t_1 + t_2) - t_1$ and player 2's payoff can be written as $\theta_2(t_1 + t_2) - t_2$. By Lemma 5, θ_1 and θ_2 are both strictly concave functions.

When in a given period τ , player 1 chooses his contribution t_1 , he has to take into account the immediate payoffs $\theta_1(t_1 + t_2) - t_1$ and the effect that his choice of t_1 will have on player 2's subsequent choices. The latter does not depend on the value of t_2 in period τ . But a higher t_2 in period τ decreases $\theta'_1(t_1 + t_2)$ by the concavity of θ_1 , so the optimal t_1 can only decrease in t_2 .

¹⁹For the Lemma to hold we only need that the fund's reward payment schemes depend only on its aggregate budget and that the benefits that it creates are concave for both players.

More formally, player 1's problem in period τ is to choose t_1 so as to maximize $h_{1,\tau}(t_1, t_2) := \theta_1(t_1 + t_2) - t_1 + \beta V_{1,\tau+1}(t_1, t_2)$, where $V_{1,\tau}(t_1, t_2)$ denotes the continuation value of the expected discounted payoffs in the remaining periods. We have $V_{1,\tau+1}(t_1, t_2) = \frac{1}{2}V_{1,\tau+1}^1(t_2) + \frac{1}{2}V_{1,\tau+1}^2(t_1)$, where $V_{1,\tau+1}^1(t_2)$ denotes the continuation value for player 1 if it is his turn again in period $\tau + 1$ and $V_{1,\tau+1}^2(t_1)$ denotes the continuation value for 1 if it is player 2's turn in period $\tau + 1$. We thus have:

$$h_{1,\tau}(t_1, t_2) := \theta_1(t_1 + t_2) - t_1 + \frac{\beta}{2}(V_{1,\tau+1}^1(t_2) + V_{1,\tau+1}^2(t_1))$$

From that we deduce that $\frac{\partial^2 h_{1,\tau}}{\partial t_1 \partial t_2} = \theta_1''(t_1 + t_2)$. By the strict concavity of θ_1 , this is negative, which implies that increasing t_2 can never strictly increase the optimal t_1 .

The proof for the claim at any Markov Perfect Equilibrium is identical. $\square$

Lemma 22. *Consider the game where there is only one combined compensation fund for both supply reduction and demand reduction. Moreover, suppose that this compensation fund uses any given budget so as to maximize global welfare (not taking into account that funding is actually endogenous). Suppose we have a Markov Perfect Equilibrium such that the Markov chain induced by it converges to a point. Then at this convergence point aggregate contributions are lower than at any Nash equilibrium of the hypothetical one shot simultaneous move game.*

Proof. Suppose we have a Markov Perfect Equilibrium and let (t_1, t_2) a point to which the induced Markov Chain converges. Suppose player i where to deviate. Using the notation from the previous Lemma, we have:

$$\begin{aligned} V_{1,\tau+1}^2(t_1) &= \theta_1(t_1 + \hat{t}_2(t_1)) - t_1 + \beta V_{1,\tau+2}(t_1, \hat{t}_2(t_1)) \\ V_{1,\tau+1}^1(t_1) &= \theta_1(t_1 + \hat{t}_2(t_1)) - t_1 + \frac{\beta}{2}(V_{1,\tau+2}^1(\hat{t}_2(t_1)) + V_{1,\tau+2}^2(t_1)) \\ h_{1,\tau}(t_1, t_2) &:= \theta_1(t_1 + t_2) - t_1 + \frac{\beta}{2}(V_{1,\tau+1}^1(t_2) + V_{1,\tau+1}^2(t_1)) \\ &\quad + \frac{\beta}{2}(V_{1,\tau+2}^1(\hat{t}_2(t_1)) + V_{1,\tau+2}^2(t_1)) \end{aligned}$$

Now using that discounted payoffs are bounded, we can continue expanding this expression indefinitely and obtain:

$$h_{1,\tau}(t_1, t_2) := \theta_1(t_1 + t_2) - t_1 + \frac{\frac{\beta}{2}}{1 - \frac{\beta}{2}}(\theta_1(t_1 + \hat{t}_2(t_1)) - t_1 + V_{1,\tau+2}^1(\hat{t}_2(t_1)))$$

Using the fact that by assumption we have $\hat{t}_2(t_1) = t_2$, we get:

$$\frac{\partial h_1}{\partial t_1} = \theta_1'(t_1 + t_2) - 1 + \frac{\frac{\beta}{2}}{1 - \frac{\beta}{2}}(\theta_1'(t_1 + t_2) - 1 + (\theta_1'(t_1 + t_2) + V_{1,\tau+2}^1'(\hat{t}_2(t_1)))\frac{d\hat{t}_2}{dt_1})$$

But by the envelope theorem we have $V_{1,\tau+2}^1'(\hat{t}_2(t_1)) > 0$ and by the preceding Lemma we have $\frac{d\hat{t}_2}{dt_1} < 0$. Thus the only way for $\frac{\partial h_1}{\partial t_1}$ to be nonnegative is for $\theta_1'(t_1 + t_2) - 1 > 0$. By symmetry we also have $\theta_2'(t_1 + t_2) - 1 > 0$. At any Nash equilibrium in the hypothetical static one shot game one of these expressions must be 0. Thus by the concavity of θ_1 and θ_2 it follows that at the MPE's convergence point $t_1 + t_2$ must be smaller than at the static Nash equilibrium. $\square$

The preceding Lemmas suggests that the rational reasoning captured by backward induction and Markov Perfect Equilibria aggravates the cooperation

problem relative to the Nash equilibrium: Players know that increasing their contributions has the negative side effect of reducing others' incentives to contribute in the future. This helps explain the pessimistic result in the following example:

Example 1. As in the numerical case computed for section 7.3.2, suppose that benefits of the two players' utility functions are quadratic and symmetric. Specifically, assume the following:

$$U_1(t_1, t_2) = 2(t_1 + t_2) - \frac{1}{2}(t_1 + t_2)^2 - t_1$$

$$U_2(t_1, t_2) = 2(t_1 + t_2) - \frac{1}{2}(t_1 + t_2)^2 - t_2$$

Moreover, suppose that the discount factor β is 1. Then the backward induction solution (computed in a Mathematica notebook downloadable [here](#)) is as follows:

$$\text{period 0: } \hat{t}_2(t_1) = \max(0, 1 - t_1)$$

$$\text{period -1: } \hat{t}_2(t_1) = \max(0, \frac{1}{2} - t_1)$$

$$\text{period -2: } \hat{t}_2(t_1) = 0$$

$$\text{period -3: } \hat{t}_2(t_1) = \max(0, 1 - t_1)$$

...

Here we see that the backward induction logic drives strategies down in period -1 . This is because players know that in period 0 their contributions will crowd out the other's contributions. In period -2 this same logic drives strategies down to 0. This means, on the other hand, that there is no more crowding out to fear when deciding in period -3 . Things are again as if this was the last period. And then a new such cycle of crumbling strategies continues.

B Mathematica notebooks

A Mathematica notebook for the numerical computations under constant elasticity specifications can be downloaded [here](#).

A Mathematica notebook for the symbolic computations used in the proof of Proposition 1 and Proposition 2 can be downloaded [here](#).

A Mathematica notebook computing the spending paths on the three approaches by the global institution at the optimal mechanism with full commitment and no borrowing or saving constraints can be downloaded [here](#).

A Mathematica notebook computing the surfaces shown in section C about the loss from misallocation can be downloaded [here](#).

C Robustness checks about the loss from misallocation

In section 4 I showed for a particular combination of elasticity estimates how welfare depends on the budget split. One takeaway was that the loss from misallocation is relatively small: as long as each of the three approaches gets at least 50% its optimal proportion of the budget, welfare losses are at most 12%. This

particular way of summarizing the “flatness of the welfare surface around the optimum” is motivated by the discussion presented in section 8 concerning the pros and cons of flexible decentralized mechanisms: We should expect countries’ allocation decisions in such mechanisms to be guided by a mixture of concern for global welfare and their own payoffs. Large fossil fuel exporters will strongly prefer money to go to the supply reduction approach since this will raise fossil fuel world market prices. Fossil fuel importers, on the other hand, will prefer money to go to the demand reduction and substitute expansion approaches. Those countries primarily concerned about climate change will prefer money to go at the margin to whatever of the three approaches is underfunded relative to the others. Thus we should expect the overall allocation to be somewhat responsive to what actually turns out to be good for global welfare. Based on this, I now assume for concreteness that for any given overall budget each of the three approaches gets at least half its optimal proportion.

Under this constraint, the worst outcome in terms of global welfare occurs when two of the three approaches each get only half their optimal proportion of the budget, with the third approach getting the rest. I refer to the three corresponding cases as “supply-side-heavy”, “demand-side-heavy” and “substitute-side-heavy”. I plot below the proportion of welfare realised under these three cases relative to the welfare that would be realised if the budget was split optimally across the three approaches. For these numerical simulations I assume constant-elasticity specifications for coal supply, energy demand and renewable energy supply. Throughout I assume that the overall budget is small. It turns out that all the results change little with the size of the budget. The plots explore the entire range of elasticity estimates that I have found in the literature, as I detail in the following subsections.

C.1 Estimates of long run price elasticities of demand for energy

Espey and Espey (2004) carried out a meta-analysis about residential electricity demand. of price and income elasticity estimates from 36 studies published over the period 1947 to 1997. The 125 estimates of long-run price elasticity fell in the range from -2.25 to -0.04 with a mean of -0.85 . All the more recent studies that I have seen have estimates falling in this range²⁰ I thus consider the range -2.25 to -0.04 in the plots shown below.

C.2 Estimates of price elasticities of supply of renewable energy

I have only found a single study, namely Johnson (2011), which gives an estimate of 2.7. In the plots shown below I consider the range from 0.1 to 3 for the price elasticity of supply of renewables.

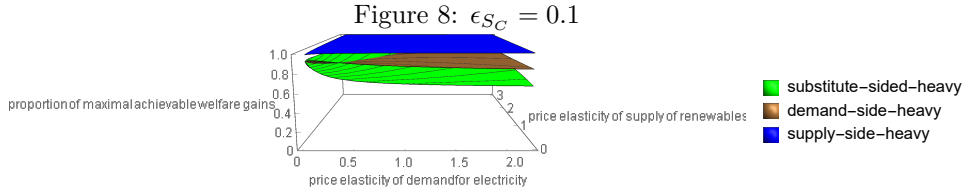
²⁰E.g. Burke & Abayasekara (2018) find -1.

C.3 Estimates of the price elasticity of supply of coal

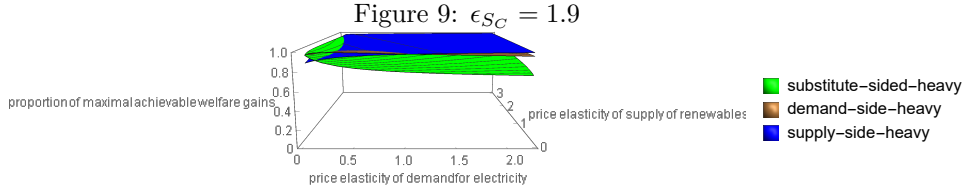
Daubanes, J., Henriët, F., & Schubert, K. (2020). note that the empirical literature on the price elasticity of coal supply—e.g., Labys et al. (1979), Beck et al. (1991), Light (1999), Light et al. (1999), and Dahl (2009)—finds estimates ranging from 0.1 and 1.9. Based on this, I consider the range from 0.1 to 1.9.

C.4 Results

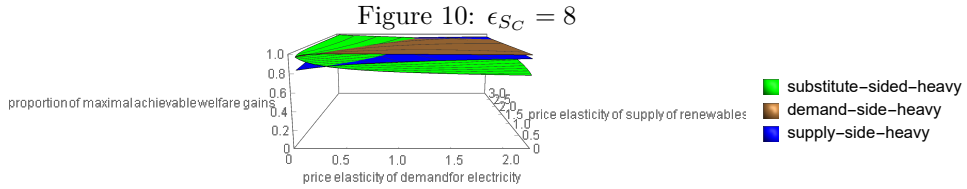
Here is the case where the price elasticity of supply of coal is 0.1:



Here is the case where the price elasticity of supply of coal is 1.9:



Whilst the estimates for the price elasticity of supply of coal range from 0.1 to 1.9, we presumably cannot rule out potentially much large value for it in the long term. For illustration, consider the case where the price elasticity of supply of coal is 8:



Overall, these results suggest that the conclusion that the welfare losses from misallocation are likely to be small is robust.

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